Conditional Trade Policy for Peace? The Economic Effects of QIZs in the Middle East

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Abstract

This paper examines how conditional trade policies with input-sourcing requirements affect firms' exports, production, and labor outcomes. I study Egypt's Qualifying Industrial Zones (QIZ) program, which grants duty-free access to the U.S. market conditional on sourcing inputs from Israel. I estimate event-study and triple-difference specifications using firm-level customs data, industrial census records, and labor force surveys.I find large and persistent export gains along both intensive and extensive margins alongside improvements in output, productivity, employment, and wages. The improvements are driven by U.S. market access rather than the quality of Israeli inputs. To map these firm-level responses into aggregate and welfare implications, I develop a quantitative heterogeneous-firm trade model with endogenous QIZ compliance, rules-of-origin costs, and productivity upgrading. The framework is used to quantify the program's economy-wide effects and conduct counterfactual analyses that vary the Israeli content requirement, highlighting the trade-off between policy conditionality and gains from market access.

Keywords: Trade policy, Place-based policies, QIZ, Industrial Policy, Development

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1 Introduction

In recent decades, there has been increased interest in place-based policies and their role in stimulating local development and job creation. At the same time, trade has long been recognized as a powerful driver of economic growth and productivity, dating back to the Industrial Revolution and even earlier. Yet the interaction between these two areas remains less well understood. What happens when a trade policy is tied to a specific location? And what are the consequences when preferential market access is granted only to firms operating in certain regions and sourcing inputs from particular partners? This paper explores this intersection by studying the economic effects of Qualifying Industrial Zones (QIZs), a policy instrument that links place-based targeting to conditional trade preferences.

Under the QIZ program, firms located in designated zones in Egypt and Jordan receive duty-free access to the U.S. market conditional on sourcing a minimum share of inputs from Israel. This design reflects economic and geopolitical objectives simultaneously, making QIZs a unique setting for analyzing how geographic targeting, trade incentives, and input-sourcing rules interact. The central research question of this paper is therefore:

How do conditional, place-based trade preferences affect firm productivity, export upgrading, and local labor-market outcomes?

The literature on place-based policies has expanded considerably in both developed and developing countries (Neumark, 2018). Evidence shows that geographically targeted interventions can raise employment, output, and productivity through agglomeration effects, although impacts vary widely with local conditions and institutional design (Lu, Wang, & Zhu, 2019; Wang, 2013; Siegloch, Wehrhöfer, & Etzel, 2025; Kline & Moretti, 2014a). In parallel, the literature on trade policy documents substantial gains in productivity and firm upgrading when barriers to trade are reduced (Amiti & Konings, 2007; Bustos, 2011). However, much less is known about trade policies that are both conditional and geographically targeted, and how they shape firms, workers, and regional economic activity.

This paper contributes to both literatures by evaluating the economic and labor-market effects of QIZs, primarily in Egypt with supporting evidence from Jordan. By examining a policy that conditions preferential market access on both location and input sourcing, the paper sheds new light on the interaction between spatial targeting, supply-chain rules, and firm behavior.

The Qualifying Industrial Zone (QIZ) agreement was first established between Jordan, Israel, and the United States in 1996 and implemented in Jordan in 1998. It designated specific regions within Jordan as QIZs, allowing firms located in these areas, to export to the United States tariff free and quota free (Pelzman, 2011). This preferential access was conditional on the use of a specified minimum share of inputs sourced from Israel. In essence, the agreement extended the benefits of the United States–Israel Free Trade Agreement to eligible firms operating within designated zones in Jordan, provided they incorporated Israeli inputs into their production process. Although it used different tools, the logic behind the QIZ agreement resembled that of the Marshall Plan, in that it aimed to stimulate development and foster economic integration among countries previously in conflict, with the expectation of generating a virtuous cycle of peace and prosperity (Bianchi & Giorcelli, 2023; Yadav, 2007). In 2004, Egypt signed a similar agreement with Israel and the US, and specific areas within it were designated as QIZs. In both Egypt and Jordan, these zones were introduced gradually over time. The key difference is that in Jordan, QIZs were mainly fenced industrial zones, while in Egypt, entire administrative regions were designated as QIZs ([Author name redacted], 2013).

When observing the aggregate evolution of trade between each of Egypt and Jordan, with both Israel and the US, we notice that both imports from Israel and exports to the U.S have increased substantially after the QIZ agreement was enforced. Was this increase in trade a consequence of the QIZ agreement or a product of international trade dynamics that are not specific to this agreement? To answer this question, I apply the synthetic control method and compare Egyptian and Jordanian exports to the United States and imports from Israel to their respective synthetic control created from a pool of other countries. The synthetic control matches the trade dynamics prior to the implementation of the QIZ agreement in each of the two countries. I find that after the implementation of the QIZ agreement, manufacturing (and more specifically textile and apparel) exports from Egypt and Jordan to the United States increased substantially compared to their synthetic counterparts. Total imports from Israel increased substantially relative to the synthetic control as well. These results suggest that the QIZ agreement might have achieved two of its objectives, stimulating trade in Egypt and Jordan and promoting regional economic integration with Israel. A closer look at Jordan, which signed a free trade agreement with the United States in 2004 that was fully implemented by 2010, we observe a sharp decline in imports from Israel after 2010, even as exports to the United States continued to grow. This suggests that Jordan's trade with Israel was not driven by deeper cultural or political ties, but was instead a pragmatic economic choice shaped by the incentives of the QIZ agreement. Once a more favorable alternative—the United States–Jordan Free Trade Agreement—became fully available, imports from Israel declined and integration as measured by trade weakened.

I further investigate the case of Egypt and compare its exports to the United States with those of its synthetic control and ask whether the observed increase can be explained solely by rising demand from U.S. consumers. To do this, I use estimates of the price elasticity of substitution for U.S. textile and apparel imports and calculate whether the tariff reductions, if fully passed through to prices, are sufficient to account for the export increase (Feenstra, Luck, Obstfeld, & Weinstein, 2018; Lashkaripour, 2022). While a demand-driven explanation accounts for a large share of the observed growth, a substantial and statistically significant portion remains unexplained. This suggests that the QIZ agreement may have also led to productivity gains, particularly through increases in total factor productivity.

Holding on to this insight, which suggests that affected industries may have experienced productivity increases, I zoom in on Egypt and investigate the effects of the QIZ agreement at the micro level. To do so, I use two identification strategies. The first strategy builds on the requirements of the QIZ agreement. To access the U.S. market under this agreement, firms are required to use a fixed share of their inputs from Israel in their production process. Although the agreement allows all firms to benefit from preferential access, most firms that actually used the policy were in the textile and apparel sector, followed by firms in the food sector. This is because the benefits of the QIZ agreement could only be fully realized by firms producing goods with otherwise high MFN tariffs, which justify the use of higher-cost Israeli inputs. A firm would only benefit from the QIZ if the tariff reduction exceeded the additional cost of Israeli inputs, and this condition was met primarily for textile and apparel products.

Using Egyptian firm-level customs data on imports and exports at the HS6 product level from 2005 to 2016, I estimate event-study models where the “event” is defined as the moment a textile firm begins importing from Israel after previously never doing so (see Alfaro-Ureña, Manelici, and Vasquez, 2022, for a similar approach). Since both data and anecdotal evidence suggest that importing from Israel is more costly than importing from other countries (Author name redacted, 2013), only firms that benefit from a sufficiently large tariff reduction have a clear incentive to do so. This makes first-time importing from Israel a reasonable proxy for entry into QIZ treatment status. I find that the exports of treated firms to the

United States rise substantially and that these increases persist in the long run. Restricting the sample to textile and apparel firms, I find that firms that begin importing from Israel not only export more to the United States but also expand exports to non-U.S. destinations relative to control firms, with effects that remain over time. I also find sizable and persistent effects of entry into importing from Israel on the number of products exported and the number of destinations served. Exploring heterogeneity across firm types, I find that earlier-treated, incumbent, and larger firms drive much of the observed effects, consistent with Egger, Nigai, and Shingal (2024).

The identification of these effects relies on the parallel-trends assumption between treated and control firms. Although this assumption cannot be tested directly, I assess its plausibility by showing that treated and control firms follow similar pre-treatment trends in all outcome variables. To further strengthen the analysis, I conduct two placebo tests. The first defines a placebo sector, for example wood and furniture, and applies the same event-study approach. The second assigns a placebo country and defines the event as beginning to import from that country rather than from Israel. In both tests, I find null effects, which reinforces the validity of the main findings. The results are also robust to a wide range of alternative specifications and robustness checks.

The second identification strategy exploits the staggered rollout of QIZs across Egyptian regions together with the fact that the policy primarily targeted the textile and apparel sectors. This setting allows me to estimate a triple-difference model, where the coefficient of interest is the interaction between time, treated-region, and treated-sector indicators. Using the available firm-level information from two industrial censuses and focusing on the most recent QIZ expansion in 2013, I find that textile firms in treated regions import more inputs, are more productive, and employ more workers, particularly men. Balance tests indicate that the pre-treatment sample is well balanced across a broad set of firm characteristics.

Applying the same empirical strategy to annual labor force surveys, I find positive effects of QIZs on wages and formality, measured by social-insurance coverage. Two advantages of the labor-force surveys are that they allow direct tests of the parallel-trends assumption and make it possible to condition on workers' occupations. The positive effects appear to be concentrated among men, while the estimates for women are considerably noisier. This pattern contrasts with evidence from countries such as Bangladesh, where the expansion of the textile industry has been shown to disproportionately benefit women (Heath & Mobarak, 2015).

Motivated by this evidence, I develop a quantitative heterogeneous-firm trade model tailored to the QIZ setting. The model features endogenous export participation, compliance with rules-of-origin through costly Israeli input sourcing, and productivity upgrading induced by access to the U.S. market. This structure allows the model to aggregate firm-level responses into economy-wide effects, quantify welfare changes, and evaluate counterfactual policy designs. In particular, I wille use the framework to trace how the gains from market access depend on the stringency of the Israeli content requirement, varying it from its baseline level in the agreement to no content requirement. While the reduced-form estimates identify large export and labor-market impacts, the model is needed to distinguish the roles of selection, compliance costs, and upgrading, and to assess how alternative rules-of-origin would reshape both aggregate welfare and the distribution of gains across firms and regions.

Beyond the literature referenced so far, this paper is closely related to three strands of research. The first is the literature on trade liberalization and trade policy, which examines how agreements that reduce trade barriers affect firms and workers (Autor et al., 2013; Opalova, 2010; Costa et al., 2016; Choi et al., 2024). A growing body of work emphasizes the role of agreement design, including rules of origin and conditional market access. Sytsma (2022) shows that relaxing rules of origin can expand firm participation in preferential trade schemes, underscoring the importance of input-sourcing constraints for understanding heterogeneous firm responses. Another related line of work uses export shocks as quasi-experiments. Lane (2025) and Barteska et al. (2025) exploit large, plausibly exogenous foreign-demand shocks to study how firms upgrade, raise productivity, and expand into new markets.

The second strand is the literature on industrial policy. Much recent work has shifted toward micro-founded evidence on targeted interventions. Lane (2024) analyzes Korea's manufacturing push and documents large and persistent firm-level upgrading effects. Manelici and Pantea (2021) study Romania's IT-sector tax exemption and show that a narrowly targeted policy can generate productivity spillovers and labor-market impacts in treated regions. These papers highlight that industrial-policy outcomes depend heavily on sectoral focus, implementation details, and existing industrial structure.

The third strand is the literature on place-based policies. Most early work focuses on developed countries (Neumark, 2015), but a growing number of studies analyze developing contexts. Wang (2013) shows that special economic zones in

China attract foreign direct investment and raise wages, while Gallé et al. (2024) find that zones in India accelerate structural transformation, especially for women. Similar studies examine zones in Vietnam and Africa (Abagna, Hornok, and Mulyukova, 2025; Treibich et al., 2025; Tafese, Lay, and Tran, 2025).

This paper contributes to all three literatures by studying a policy that is simultaneously a place-based intervention, a form of industrial targeting focused on textiles and apparel, and a conditional trade policy tied to sourcing requirements and preferential access to the U.S. market.

A comprehensive overview of the history of QIZs in Egypt and Jordan is provided by a U.S. Congress report (Author name redacted, 2013). Several papers have studied the effects of QIZs on trade and employment in Egypt (Yadav, 2007; Nugent and Abdulatif, 2010) and in Jordan (Warad, 2010; Pelzman, 2011; Saif, 2006). However, these studies rely on aggregate data and do not attempt to causally identify the effects of QIZs. This paper contributes to the literature by using microdata from Egypt and applying econometric methods to estimate the causal impact of QIZs on firms and workers. To the best of my knowledge, this is the first study to apply microeconometric methods, such as event-studies and triple-difference designs to examine the effect of a place based conditional trade policy, like the QIZ, on firm-level and labor outcomes.

The remainder of the paper is organized as follows. Section 2 provides background on the institutional context and history of the QIZ agreements. Section 3 describes the data. Section 4 outlines the empirical strategies. Section 5 presents descriptive statistics and the SCM results from the aggregate trade data. Section 6 presents the event-study firm level results from customs data. Section 7 and 8, presents results from the industrial censuses and labor force surveys by employing a triple differences design. Section 9 introduces the quantitative heterogeneous-firm model and uses it to quantify welfare effects and conduct counterfactual analyses that vary the Israeli content requirement. Section 10 discusses the results, and section 10 concludes.

2 Institutional Context and History

The Arab–Israeli conflict in the 20th century involved multiple Arab countries engaging in wars with Israel in an effort to secure the rights of Palestinians to return to their homes and establish an independent state. During the conflict, particularly after the 1960s, the United States aligned itself with Israel, while the Soviet

Union supported the Arab states involved (Khalidi, 2020). In 1979, Egypt and Israel signed the Camp David Peace Accords, which formally ended the state of war between them following the October War of 1973. Later, in 1994, Jordan and Israel signed the Wadi Araba Peace Treaty, following the 1993 Oslo Accords between the Palestine Liberation Organization and Israel. Both peace treaties were brokered by the United States, which had a strategic and geopolitical interest in aligning these parties more closely within its sphere of influence.

The conflict left the economies of the arab states weaker and inflicted significant economic and political costs on them. To further consolidate this regional alignment and strengthen the cold peace between its allies, the U.S. offered Jordan and Egypt the QIZ agreement in 1996. The agreement states that upon the agreement of Israel with its Arab partner (Egypt or Jordan), certain regions would be designated as QIZs and would benefit from special trade privileges in the U.S. Firms producing within these QIZ regions would be eligible to export to the U.S tariff free and qouta free upon sourcing a fixed amount of inputs from the country itself, and from Israel. Moreover, a substantial transformation of the product must take place within the QIZs. For both Jordan and Egypt, country input shares should total 35% of the appraised value of the product upon entry to the U.S. For Jordan, of these 35 %, 8% should come from Israel, 11.7 % from Jordan itself, and the remaining 15.3 % from any combination of Jordanian QIZ, Israel, the U.S, and the West bank and Gaza strip (Author name redacted, 2013). For Egypt, of these 35 %, 10.5% should come from Israel, another 10.5 % from Egypt itself, and the remaining 15.3 % from any combination of Jordanian QIZ, Israel, the U.S, and the West bank and Gaza strip (Author name redacted, 2013).

Egypt's entry into the agreement came in 2004, much later than Jordan's participation in 1996. Initially, Egypt was skeptical of the QIZ arrangement, in part because it was already benefiting from the World Trade Organization (WTO) Agreement on Textiles and Clothing (ATC) (WTO, n.d.). However, the expiration of the ATC in January 2005 pushed Egypt to explore alternative mechanisms to maintain preferential market access, including the QIZ agreement (Author name redacted, 2013). The observed success of the QIZs in Jordan, particularly in boosting exports and creating new jobs, made the idea more appealing, provided it did not provoke public backlash. To preempt such concerns, the Egyptian government insured the support of the Islamic institution of Al-Azhar before formally entering the agreement (Raya, 2006).

QIZs were added in both Jordan and Egypt successively in a staggered manner.

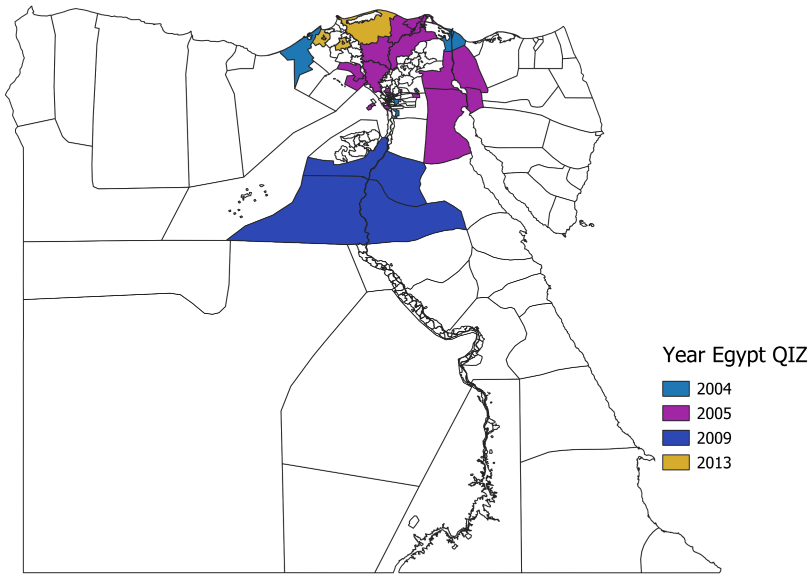


Figure 1: Map of Egyptian QIZs

A key difference between the two countries lies in the spatial design of the zones. In Jordan, QIZs were typically fenced industrial areas located in rural regions. In contrast, Egypt designated entire administrative districts as QIZs, meaning that any firm operating within those districts could potentially benefit from the agreement. Figure 1 shows the geographic distribution of QIZ regions across Egypt.

Another important aspect is that, initially, firms located within Egyptian QIZs needed individual approval from a joint QIZ committee in order to qualify for the agreement. This committee included representatives from Egypt, Israel, and the United States (QIZ Egypt, n.d.). In practice, these approvals were granted almost exclusively to textile and apparel firms. Even after the 2013 liberalization, when the United States confirmed that all firms and facilities, both existing and future, located within designated QIZ districts would automatically qualify, participation remained concentrated in the textile and apparel sector. Over time, firms in the food processing sector also began to join the program in increasing numbers, as shown in Figure 2. The reason why firms from other sectors have not participated is that U.S. tariffs on their products tend to be alreadz relatively low, and as a result, the cost savings from tariff exemptions under the QIZ agreement are not enough to offset the higher input costs associated with sourcing from Israel (Author name redacted, 2013). Industry accounts suggest that for firms to benefit meaningfully from the agreement, tariff rates must be at least 10 percent in order to create a sufficient margin to compete with lower-cost producers such as those in China and Bangladesh(ibid).

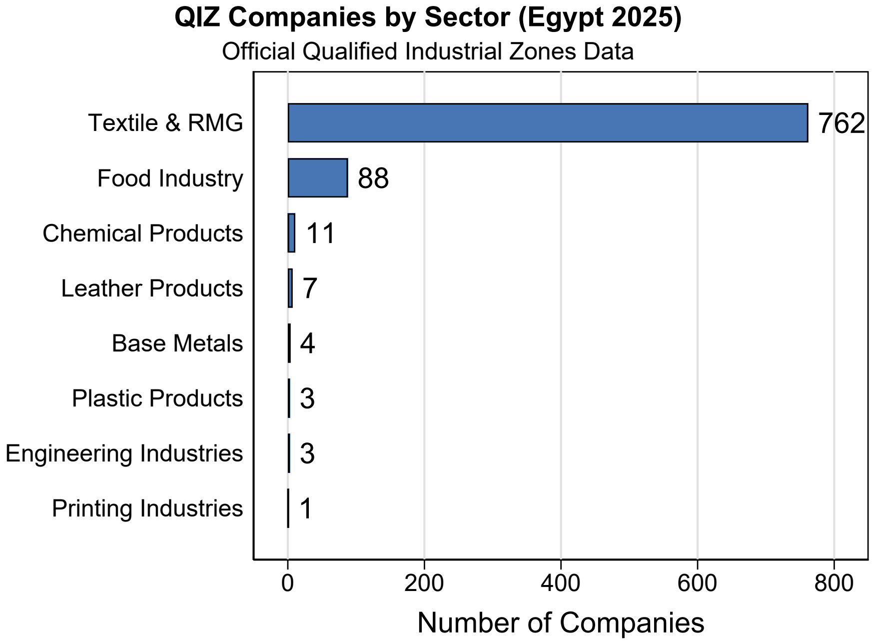


Figure 2: QIZ Companies by Sector

The daily administration of the program is overseen by an official unit within both Egypt and Jordan, known as the QIZ Unit. These units are responsible for registering firms, issuing and renewing certifications, and facilitating all administrative procedures related to participation in the program. In addition, according to the QIZ protocols, a joint committee composed of Israeli and Egyptian or Jordanian representatives is required to meet on a quarterly basis to coordinate and oversee the implementation of the agreement (QIZ Egypt, n.d.). However, since Jordan signed its own Free Trade Agreement with the United States in 2004—which was fully implemented by 2010 to cover all products, including those produced in QIZs—the QIZ agreement gradually became obsolete. Unlike the QIZ framework, the FTA does not impose input sourcing requirements, making it a more flexible and attractive option for Jordanian exporters (Saif, 2006, Areiel & Yahn, 2019).

The expansion of QIZs in both Jordan and Egypt is a trilateral decision. This means that Egypt or Jordan must first reach an agreement with Israel on designating a new QIZ, after which the United States must also approve the addition. U.S. approval is typically granted through the Office of the United States Trade Representative, which oversees trade agreements on behalf of the U.S. government.

3 Data

Data on QIZs

I compiled data on QIZs in Egypt from multiple sources. The primary source is the U.S. Federal Register, which documents each new addition to the QIZ program and records the governorate in which the zone is located as well as the date of its official designation. In the main analysis, I treat the governorate as the unit of treatment, since the remainder of the data is either representative at this level or only available at the governorate level. A governorate is therefore coded as treated if at least one area within its boundaries contains a QIZ. In addition, I gathered more granular information from the website of the Egyptian QIZ Unit, which provides the exact district locations of QIZs. I use this district-level data in a some of the robustness checks.

Trade Data

To explore what happens at the aggregate trade level and to compare the performance of the textile and apparel industry in these countries to a synthetic control before and after the introduction of QIZs, I use export and import data from UN Comtrade covering the period from 1990 to 2022. Specifically, I use the value of textile and apparel exports to the US and the values of import from Israel for all countries in the dataset. I exclude the US, European countries, Australia, New Zealand, Japan, and Canada from the donor group and hence effectively restrict the sample to the set of developing countries which are closer economically to Jordan and Egypt. Since the synthetic control method does not work if missing values exist, I exclude any country with missing values for years prior to the QIZ introduction year. I also interpolate any missing values post QIZ years for the same reason.

Firm Level Data

To explore what happens to firms that begin importing from Israel after previously not doing so, I use firm-level export and import data from 2005 to 2016 compiled by the General Organization for Export and Import Control (GOEIC). I get access to this data through the Economic Research Forum (ERF). The data includes the values of import and export by source countries and destination markets at the

HS6 product level for the universe of Egyptian firms. By merging the export and import datasets through firm identifiers, I am able to track changes in firm behavior following the initiation of imports from Israel. Additionally, I can distinguish between incumbent firms and new entrants. I define new entrants as firms that do not appear in the export data in 2005, the first year of observation.

To explore how production and labor adjust at the firm level, I use two waves of the Egyptian industrial census from 2012 and 2017, accessed through the ERF. These data provide detailed information on firm production, employment, and trade, which allows me to evaluate how firms in regions included in the 2013 expansion of QIZs in Egypt change relative to firms in regions that were never included. A key limitation of this dataset is that firm location is reported only at the governorate level (similar to the state level in the United States). While some QIZs are designated at the governorate level, others are defined at the district (county) level. Because of this data constraint, I classify a governorate as treated if any area within it receives QIZ designation.

Labor Data

To explore how QIZs affect workers differentially, I use the annual Labor Force Surveys from the Egyptian Statistical Office (CAPMAS) covering the years 2006 to 2017. Since the first survey year available to me is 2006, I exclude all governorates that were treated in 2004–2005 and focus on the 2009 and 2013 expansions instead. The surveys are representative at the governorate level. I focus on wages and social-insurance coverage as outcome variables and examine how these outcomes evolve by gender for individuals working in the textile industry after a region receives QIZ status.

4 Empirical Strategies

To identify the causal impact of QIZs, I use three different empirical strategies.

4.1 SCM: Aggregate Effects on Trade

To examine the aggregate trade effects of QIZs, I use the Synthetic Control Method (SCM) to compare the evolution of textile and apparel exports to the United States and imports from Israel for Egypt and Jordan relative to a counterfactual which matches Egyptian and Jordanian outcomes in years prior to the QIZ agreement.

This approach allows me to construct a counterfactual trajectory of what trade would have looked like in the absence of the agreement. The method is particularly suited to our setting given the focus on a single treated unit and the clear start of intervention years: 1998 for Jordan and 2005 for Egypt.

I compile bilateral trade data from UN Comtrade for the years 1988 to 2022, focusing on exports of textiles and clothing to the U.S. and imports from Israel. The analysis is performed separately for each country and trade flow. For consistency, I restrict the sample to countries that have complete data on trade in these sectors during the pre-treatment period (1994–2004 for Egypt and 1990–1998 for Jordan). Countries with missing pre-treatment data are dropped, while missing values in the post-treatment period are linearly interpolated. To minimize confounding from other preferential trade agreements, I exclude developed economies and other countries with advanced bilateral trade privileges with the United States (e.g., Mexico, Canada, Korea).

Let J denote the number of control countries. The goal is to select weights $W = (w_2, ..., w_{J+1})'$, such that the weighted average of the control units best approximates the pre-treatment characteristics of the treated country. Let X_1 be a $K \times 1$ vector of pre-treatment trade outcomes for the treated country and X_0 the corresponding $K \times J$ matrix for the donor pool. The vector W^* is chosen to minimize the distance $(X_1 - X_0W)'V(X_1 - X_0W)$, where V is a positive semi-definite diagonal matrix that reflects the relative importance of predictors.

The estimated treatment effect in year t is the difference between the actual outcome for the treated unit and the synthetic control:

$$\hat{\alpha}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt},$$

where Y_{1t} is the trade outcome for the treated country and Y_{jt} is the outcome for control country j in year t.

I construct the synthetic control using a combination of export and import values in the years leading up to treatment (for example, 1994–2005 for Egypt) as predictors. Inference is based on placebo tests, in which the same model is estimated for each untreated country under the assumption that it was treated in the same year. The distribution of placebo effects is then used to assess the likelihood that the observed treatment effect could have occurred by chance.

This analysis is intended to provide descriptive and indicative evidence at the aggregate level.

4.2 Firm-Level Event Study: Entry into QIZ Treatment

To estimate the impact of entering the QIZ agreement at the firm level, I exploit detailed administrative data on the universe of Egyptian exporters and importers between 2005 and 2016, disaggregated by HS6 product and partner country. I define a firm as treated if it begins importing from Israel in year t after having no recorded imports from Israel in all years prior to t. Firms that never import from Israel over the entire sample period form the control group.

Restricted Sample

Restricting the sample to firms in the textile and apparel sector—the main beneficiaries of the QIZ agreement—I estimate the following event-study model:

$$Y_{it} = \alpha_i + \omega_t + \sum_{k=C_0}^{C_X} \theta_k D_{it}^k + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the outcome of interest (e.g., exports to the U.S.), α_i are firm fixed effects, ω_t are year fixed effects, and D_{it}^k are event-time dummies indicating k years before or after firm i begins importing from Israel. The omitted category is the year immediately prior to the treatment event. Standard errors are clustered at the firm-year level.

This specification allows us to observe the dynamic effects of entering QIZ treatment over time. I show that exports to the U.S. increase significantly following entry, and that this effect persists in the long run. The event of importing from Israel acts as a strong and credible proxy for participation in the QIZ program, given the Israeli input requirement embedded in the agreement and the higher cost of sourcing inputs from Israel relative to other suppliers.

Sectoral Comparison and Heterogeneity Analysis

I then broaden the analysis to include firms from other manufacturing sectors to assess whether the observed effects are specific to textiles or indicative of wider patterns. Specifically, I compare textile firms that begin importing from Israel with firms in other sectors, while controlling for those that experience similar importing from Israel events. I exclude firms in the food and chemical sectors, as these industries also contain companies registered under the QIZ program.

In this specification, I include sector-by-year fixed effects and interact the event-time dummies with a binary indicator for textile firms. I thus estimate:

$$Y_{ist} = \alpha_i + \omega_t + \lambda_{st} + \mathbf{1}_{\text{Textile}} \cdot \sum_{k=C_0}^{C_X} \theta_k D_{it}^k + \varepsilon_{ist}, \tag{2}$$

where λ_{st} are sector-by-year fixed effects and $\mathbf{1}_{\text{Textile}}$ equals one for firms in the textile and apparel sector. This setup captures how treated textile firms differ in their export dynamics compared to untreated firms in unrelated sectors.

Standard errors are two way clustered at the firm level ane yeal to account for correlated shocks within firms through time. While this approach could introduce some attenuation due to unrelated sectoral shocks during the event window, it provides a conservative test of whether the export gains following Israeli input sourcing are concentrated among QIZ-eligible firms.

In another specification, i intereact the event dummies with all sectors of firms which show up in the list of registered firms. Hence, in this specification I look at the outcomes of textile, food, and chemical firms which experience an event relative to all other firms.

4.3 Triple-Difference Design Using Industrial Census Data

To assess how QIZs affected local firm outcomes through regional labor demand channels, i leverage the staggered geographic expansion of QIZs across Egypt and estimate a triple-difference (DDD) specification using firm-level census data from 2012 and 2017. The 2013 expansion targeted several new governorates—most of them less developed—creating variation across sector, region, and time that I use for identification.

I restrict the analysis to the textile and apparel sector, which was the primary target of the agreement, and estimate the following specification:

$$Y_{igst} = \beta_1 \, \text{Textile}_s \times \text{TreatedGov}_g \times \text{Post}_t + \gamma X_{igst} + \omega_g + \theta_t + \phi_s t + \psi_{et} + \epsilon_{igst} \tag{3}$$

where Y_{igst} is the outcome of interest (e.g., employment, wages) for firm i, in governorate g, sector s, and year t. The coefficient β_1 captures the differential change in the outcome for textile firms located in treated governorates after the QIZ expansion, relative to other firms.

X_{igst} includes firm-level controls such as age. Fixed effects include governorate

(ω_g), year (θ_t), and sector by year ($\phi_s t$) fixed effects to control for time-varying compositional shifts. Standard errors are clustered at the governorate level.

Since only two census waves are available, I cannot conduct test for pre-trends. However, I implement a comprehensive balance check on pre-treatment observables (2012) to assess the similarity of treated and untreated firms across firm characteristics, input use, financial variables, and exposure to government programs. The results show no systematic differences, supporting the credibility of the identifying assumptions.

4.4 Triple-Difference Design Using Labor Force Survey Data

To examine how QIZ exposure differentially affects workers across genders, I use the annual Labor Force Surveys from 2006 to 2017, which are representative at the governorate level. I thus exploit the 2009 and 2013 expansions of QIZ areas and estimate the same triple-difference specification as in the firm-level analysis, now applied to individual labor market outcomes such as wages, hours worked, and social insurance status. The identification relies on parallel trends holding. I test for the existence of pretrends by estimating a dynamic version of (3) and looking at labor outcomes of treated vs untreated workers prior to QIZ expansions.

5 Synthetic Control Method Results

In this section, I present the baseline estimation results corresponding to the SCM empirical strategies outlined in Section 4. I begin by analyzing how the implementation of QIZ agreements affected aggregate trade patterns in Egypt and Jordan using a synthetic control approach (Section 5.1). I then assess whether the change in exports could be explained by demand side effects alone. Examining textile and apparel exports in Jordan and Egypt to the US and their imports from Israel, I find a large increase in the amount after both implemented the QIZ agreements in 1998 and 2004 respectively.

5.1 SCM: Egypt

Figure 3 compares Egyptian textile and apparel exports to the US, to its synthetic control. I observe a substantial increase in the value of textile and apparel exports from Egypt to the United States following the initial rollout of QIZs in late 2004. By 2011, exports had risen by approximately $ 1 billion. Interestingly, the increase

Figure 3: Egyptian Textile and Apparel Exports to the US

does not occur immediately relative to the synthetic control but exhibits a two-year lag, after which exports grow sharply. This delayed response can be explained by the expiration of the extension of the Multi-Fiber Agreement (MFA) in 2005, which previously imposed export quotas on each country (WTO, n.d.). Firms that were already exporting textiles and apparel to the United States gradually shifted to using the QIZ agreement as an alternative mechanism after the removal of MFA quotas. In addition, the lag reflects the time required for find Israeli suppliers, complete bureacratic requirements, and find buyer from the United States to export.

Figure 4 displays the total value of Egyptian imports from Israel alongside a synthetic control. We can see a sharp increase in imports after 2004, following the introduction of QIZs, relative to the synthetic counterpart. Similar to the export pattern, the rise in imports exhibits a two-year lag before accelerating, coinciding with the surge in Egyptian textile and apparel exports to the U.S. This reinforces the interpretation that the export growth was driven by the QIZ agreement. At their peak in 2012, imports from Israel reached approximately \$ 100,000—an increase of about \$ 85,000 from 2004. Notably, once imports surpass the synthetic control, they remain consistently higher. Given that Egyptian firms were required to source at least 10.5 % of inputs from Israel, the ratio of the increase in U.S. textile exports to the increase in imports from Israel is close to this mandated share, providing further evidence that these trade patterns were indeed a direct result of the QIZ agreement.

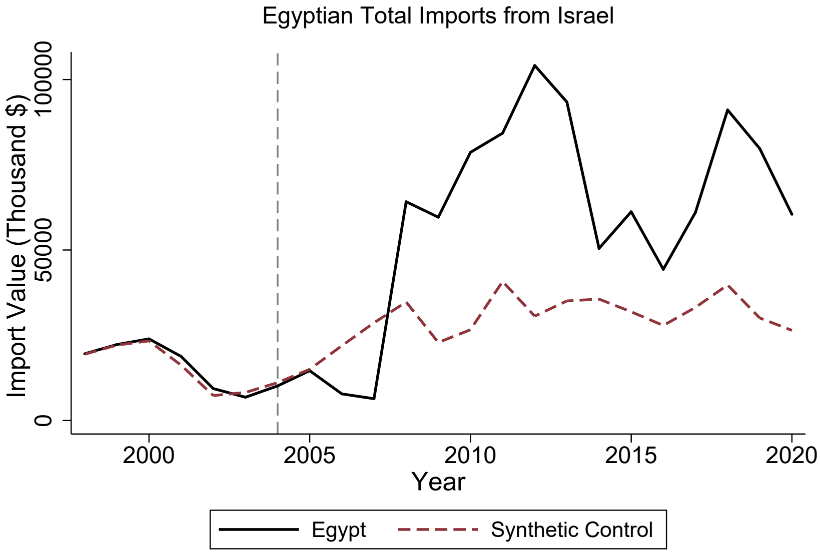


Figure 4: Egyptian Textile and Apparel Imports from Israel

Demand-Side vs. Supply-Side Forces Behind Egypt's Export Boom

To assess whether the observed surge in Egyptian apparel exports to the U.S. can be explained purely by tariff-induced demand shifts, I conduct a back-of-the-envelope calculation. The average tariff rate on textile and apparel products according to WITS (AHS weighted average, 2003) was 9.77%, hence the QIZ agreement eliminated tariffs of roughly 9.77%, reducing the relative price of Egyptian goods from 1.0977 to 1.00. This corresponds to:

$$\Delta \ln P = \ln\left(\frac{1.00}{1.0977}\right) \approx -0.0932.$$

Using an estimate of the price elasticity of substitution for textile and apparel consumers of the U.S., $\sigma = 6.7$, from Broda and Weinstein (2006), the predicted import response of U.S. consumers is:

$$\Delta \ln M = -\sigma \cdot \Delta \ln P = -6.7 \times (-0.0932) \approx 0.625,$$

which implies a multiplicative increase of:

$$e^{0.625} \approx 1.87.$$

Thus, if tariff reductions were fully passed through to prices and no other forces were at play, U.S. consumers would demand more Egyptian textile goods, and Egyptian apparel exports would be expected to increase by a factor of 1.87. Starting from a low point of \$129 million in 2007, this predicts a post-treatment level of:

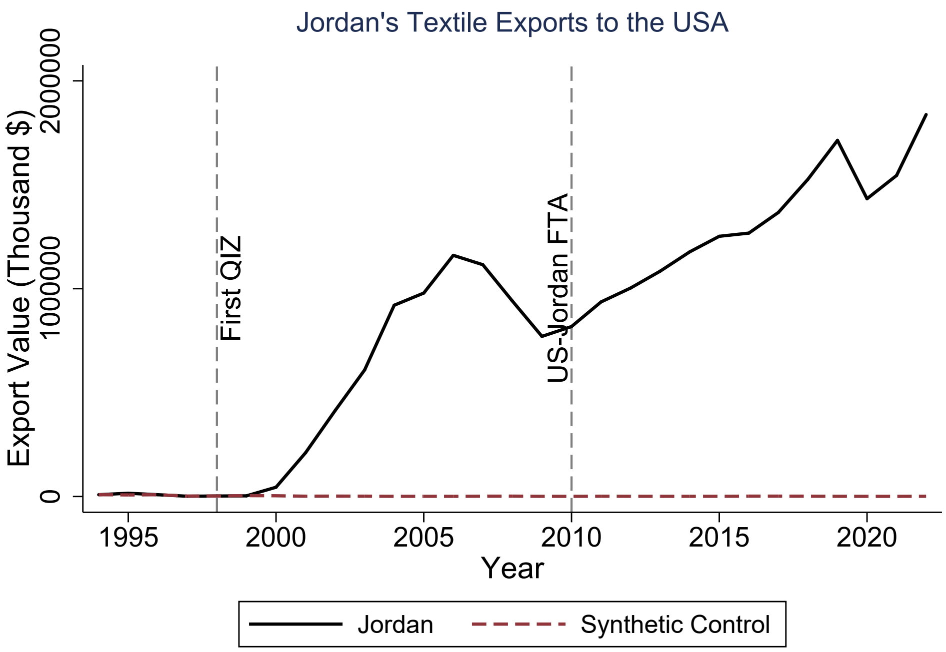
$$\hat{M} \approx 1.87 \times 129 \approx \$242 \text{ million.}$$

In reality, exports reached about \$1,105 million at their peak, an 8.6-fold increase relative to the 2007 low. This suggests that tariff-induced demand explains less than half of the actual growth. If I repeat the same exercise using the 2004 export level of \$216 million as the baseline, the predicted level would be about \$403 million, which still falls short of the observed value.

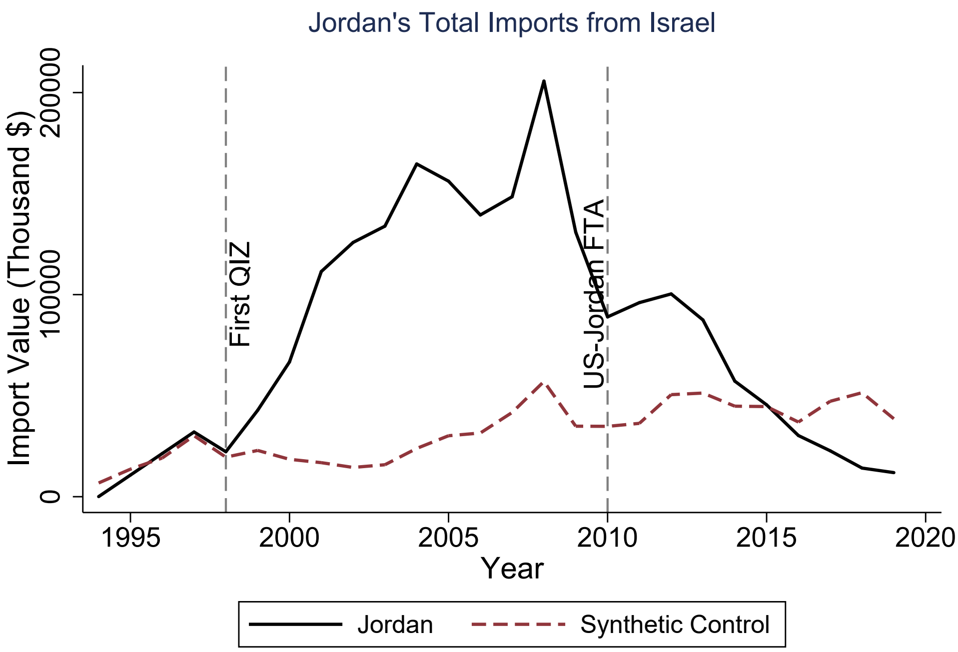
The remaining gap likely reflects supply-side forces such as capacity expansion, learning effects, fixed cost effects, and agglomeration effects within QIZ regions. Diversion of exports from other countries to the U.S. is also a possible explanation. However, as we will see in the event studies, exports to other countries increase rather than decrease for treated firms, which points to a supply-side productivity effect rather than trade diversion.

5.2 SCM Jordan

Figures 5 (a) and 5 (b) present the same export and import dynamics as Figures 3 and 4, but for Jordan. The pattern is similar. Exports to the U.S. increased by about \$ 1.1 billion by 2006, eight years after the first QIZ was designated in 1998 (Figure 5 (a)). The textile and apparel sector was very small before the QIZ agreement, which explains why the synthetic control remains flat and close to zero during the pre-treatment period. Figure 5(b) shows a corresponding rise in imports from Israel after 1998, peaking at approximately \$ 200 million. However, these imports decline sharply once the U.S.–Jordan Free Trade Agreement (FTA) is fully implemented, eventually falling below the synthetic control by 2016. This pattern suggests that the FTA offered a more favorable economic arrangement for firms, as it eliminated the costly Israeli input requirement imposed by the QIZ framework. Once the FTA applied fully to the textile and apparel sector in 2010, firms switched to the more profitable option. This implies that the regional integration fostered by the QIZ agreement was not structural but rather a second-best solution that unraveled when a more profitable alternative became available.



(a) Jordan's Textile Exports to the U.S.



(b) Jordan's Imports from Israel

Figure 5: Jordan's Trade Dynamics After QIZ Implementation

6 Event Study Results

Using Egyptian firm-level customs data, I estimate event study models as described in Section 4.2 to examine the dynamic effects of QIZ treatment on textile firm exports. To assess the robustness of my findings, I conduct two placebo tests. I further investigate whether the observed effects are driven by an overall expansion of trade or by a reallocation of exports across destinations. Finally, I explore heterogeneity in the treatment effects by distinguishing between incumbent and new entrants, as well as between early- and later-treated firms.

6.1 Textile Only Sample

I begin by estimating Equation (1), where the dependent variable is the value of firm-level exports to the U.S. As described in Section 4.2, an event is defined as the year in which a firm starts importing from Israel after not importing in the previous year. To visualize the events, Figure 6 shows the number of textile firms that experienced an event in each year of our dataset. I observe a clear spike in the years following each QIZ expansion. Although the dataset only covers 2005 to 2016 and does not include the first expansion year of 2004, there is a noticeable increase in 2006, which reflects the impact of the 2004 and 2005 expansions. A similar pattern appears in 2010 following the 2009 expansion, and again in 2015 after the 2013 expansion. Over time, the number of firms experiencing an event decreases, as earlier treated regions had more established textile and apparel firms, while later treated regions were generally less developed in terms of textile industry presence (Capmas, 2004) Figure 7 reports the results when restricting the sample to textile and apparel firms. I find no differential pre-trends between the treatment and control group, as pre-event coefficients are not statistically different from zero. On average, exports to the U.S increase by approximately $0.5 million per firm immediately after the event, although this short-term effect is not statistically significant at the 95% confidence level. In the long run, however, the effect becomes both substantial and persistent: five years after the event, treated firms export about $2 million more to the U.S. than before, and this increase remains even after eight or more years. Although the results are positive even 8 years after the event, we can notice a decrease in the coefficient of later year relative to the first 5 years after the event. This relative decrease in the coefficients is probably driven by firm exits which are captured an export value of 0 in the data. Since trade data often contains a large number of zeros in firm-level exports, using OLS on export values as the dependent variable

Figure 6: Distribution of Events for Textile Firms per Year

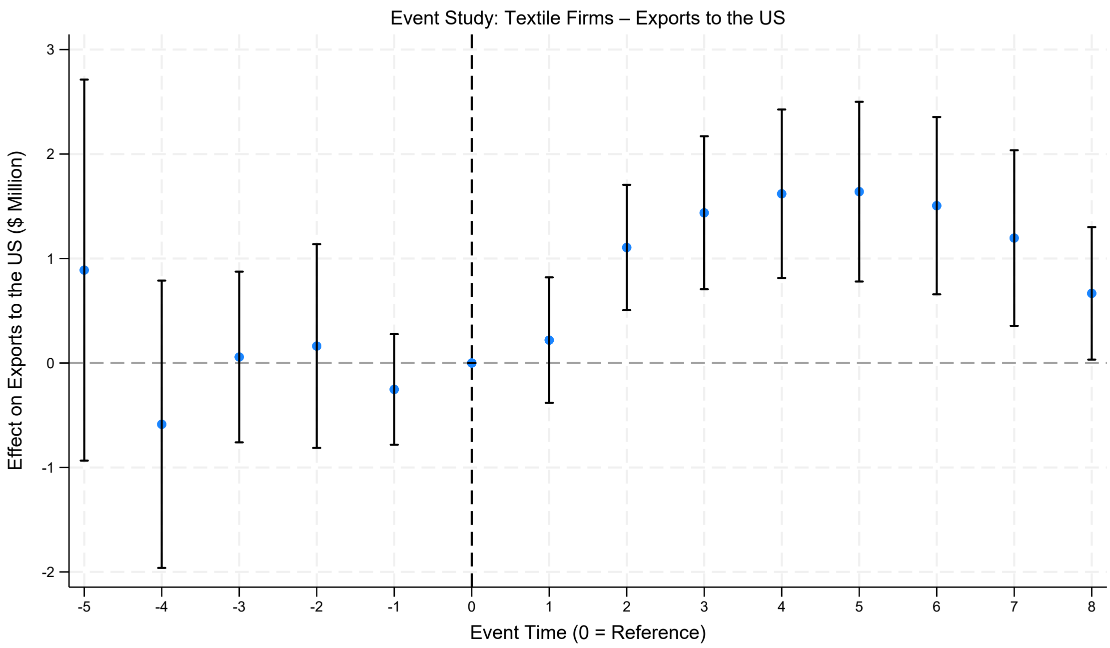


Figure 7: Event Study showing the dynamic effects of QIZs on exports to the US for Textile and Apparel Firms (OLS)

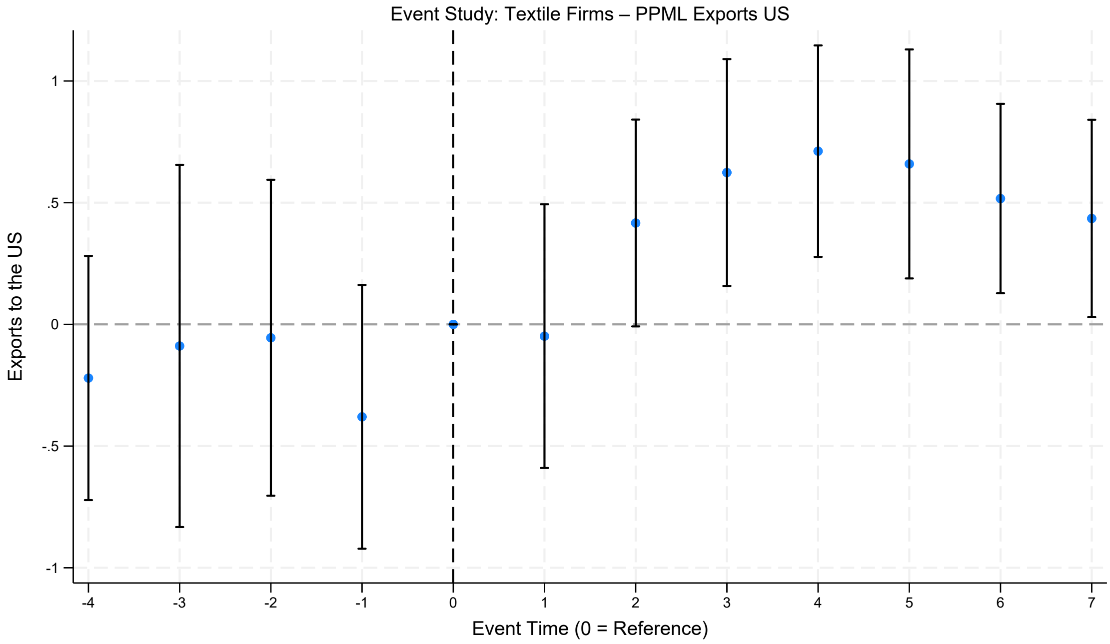


Figure 8: Event Study: Effects of QIZ on exports to the US for Textile and Apparel Firms (PPML)

can lead to biased estimates due to heteroskedasticity and the censoring problem (Santos Silva and Tenreyro, 2006). To address this issue, I estimate a Poisson pseudo-maximum likelihood (PPML) model, which accommodates zero trade flows and provides consistent estimates under heteroskedasticity. Figure 8 presents the results from estimating Equation (1) using the PPML estimator. A similar pattern emerges as in the OLS results: there are no pre-trends when observing 4 years prior to the event. Two years after the event, exports to the U.S. begin to rise, and the increase becomes economically and statistically significant in the medium and long run. By five years after the event , treated firms export substantially more to the U.S., corresponding to 100 % increase in exports, and this effect persists even seven years post-treatment. This implies that once a firm starts importing from Israel to satisfy the requirements of the QIZ agreement, it indeed increases its exports to the U.S on average. Figures A1 in Appendix A, shows similar results when taking the log of the value of Us exports + 1 as the dependent variable.

Robustness

To assess the robustness of our findings, I conduct two placebo tests. The first modifies the definition of the event by considering firms that begin importing from a neighboring country such as Sudan or Saudi Arabia after previously not importing from that country, instead of Israel. The second placebo test focuses on different

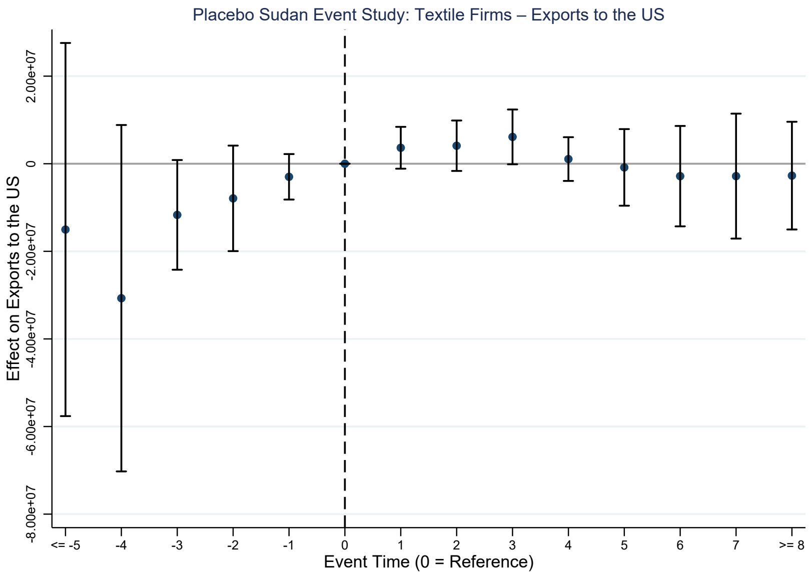


Figure 9: Event study estimates using Sudan as a placebo country.

non treated sectors, e.g. the wood and furniture sector, and examines the export behavior of firms in this sector after they start importing from Israel. If these placebo exercises do not yield effects comparable to those observed in our baseline analysis, this would provide reassurance that the estimated impacts are driven by the QIZ agreement rather than by unrelated factors.

Figure 9 presents the results for the first placebo test. In this exercise, I redefine the event as the year in which a firm begins importing from Sudan rather than from Israel and reestimate the model using OLS. The results show no evidence of differential pre-trends and, importantly, no significant effect of the placebo event on textile exports to the U.S. after treatment. This strengthens the interpretation that the event definition captures the causal impact of the QIZ agreement on the textile and apparel sector, rather than picking up unrelated changes in trade patterns.

Figure A2 in Appendix A reports the results of a placebo exercise using Saudi Arabia as the alternative source country. The overall pattern is similar to that observed for Sudan. However, in this case, I detect some pre-trends and a short-run increase in exports following the placebo event. This likely reflects a correlation between growing U.S. exports and higher imports of energy-related products from Saudi Arabia during that period. Taken together, these results support the credibility of our baseline event-study design, indicating that the main estimates capture the causal effect of QIZ participation on the textile and apparel sector rather than spurious correlations.

Moving to the next placebo test, I re-estimate the event-study model by restricting the sample to firms in a placebo sector—specifically, the wood and furniture

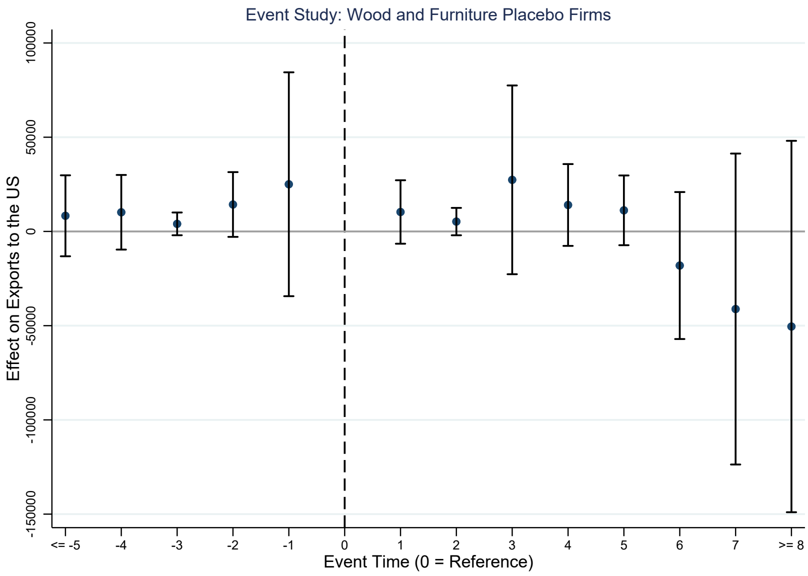


Figure 10: Placebo Sector Event Study: Wood and Furniture (OLS)

industry. Figure 10 reports the OLS estimates. Consistent with expectations, there are no significant pre-trends and no discernible post-event effects. I then repeat this exercise for every sector and plot the resulting average treatment effects in Figure A3 in Appendix A. Across all placebo sectors, I find no significant positive effects of first importing from Israel on subsequent exports to the United States. The only exceptions are textile and apparel firms and food manufacturing firms, which together account for more than 99% of all QIZ-registered firms.

These results imply that the surge in exports to the United States is not driven by the quality of Israeli inputs, otherwise, similar export responses would appear in non-QIZ sectors as well. Instead, the effects are concentrated in precisely the sectors eligible for and registered under the QIZ scheme, indicating that access to the U.S. market through the QIZ agreement is the key mechanism. Overall, these placebo tests reinforce the credibility of the main findings: only QIZ-eligible sectors exhibit treatment effects, while unrelated sectors display no comparable patterns.

6.2 All Firms Sample

To assess whether QIZ-treated textile firms experience different export responses than other firms, I estimate Equation 2, which interacts the event variable with an indicator for textile firms. Figure 11 plots the corresponding interaction effects. The results indicate that importing from Israel generates substantially different impacts for textile firms in both the short and long run.

Textile firms which are eligible for and actively use the QIZ agreement—exhibit

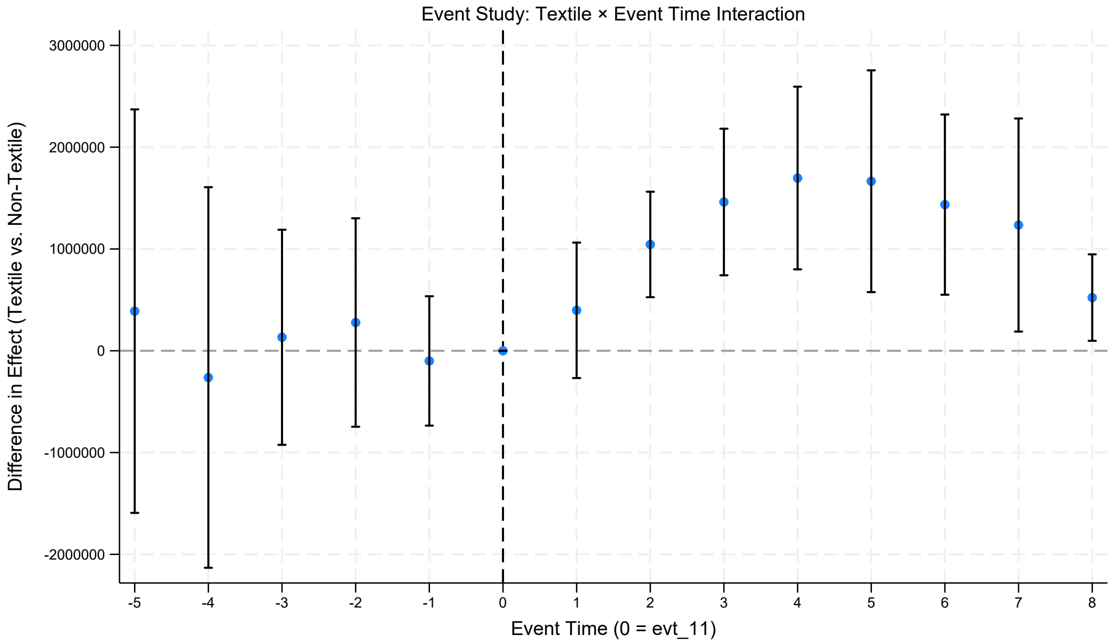


Figure 11: Event-study interaction between textile firms and QIZ treatment. OLS estimates using export values to the United States as the dependent variable.

markedly larger increases in exports to the United States compared to all other exporting firms. In other words, access to the U.S. market through the QIZ framework appears to have enabled textile firms to significantly outperform non-textile firms along the export margin.

6.3 Second Stage Effects: Other Outcome Variables

In this section, I examine how the QIZ agreement affected additional outcomes for treated firms. The increase in exports to the United States can be viewed as a first-stage outcome, since expanding access to the U.S. market was the primary objective of the QIZ agreement. To explore potential second-stage effects, I focus on three additional variables that can be constructed from the trade dataset. First, I assess whether treated firms increased their exports to non-U.S. destinations following treatment. Second, I examine whether treatment led firms to expand the range of products they exported. Finally, I analyze whether treated firms increased the number of foreign destinations to which they exported.

Non-U.S. Exports: Expansion or Reallocation?

I examine whether the increase in exports for QIZ-treated firms is driven by a reallocation of exports from other countries to the U.S. To do this, I re-estimate equation

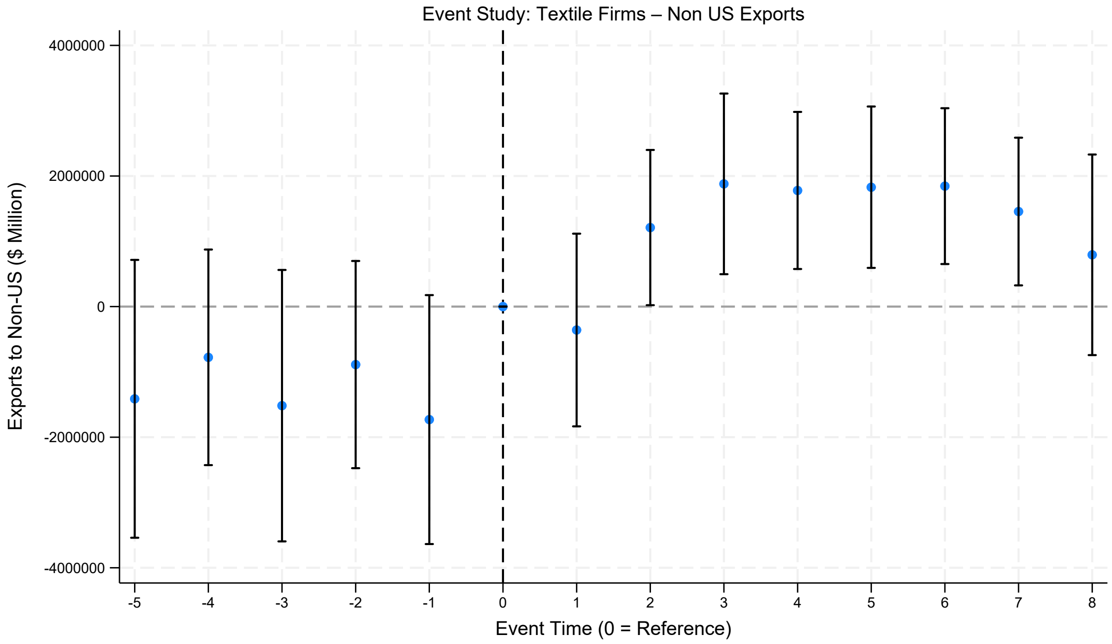


Figure 12: Event-study estimates of the dynamic effects of QIZs on exports to all non-U.S. destinations for textile and apparel firms. Coefficients are OLS estimates using export values as the dependent variable.

(1) using total non-U.S. exports as the outcome. Figure 12 plots the dynamic effects of a firm starting to import from Israel on total non-U.S. exports, relative to firms that do not import from Israel. As apparent in the figure, there is a strong and persistent positive effect of QIZ treatment on the value of non-U.S. exports exhibited by treated Egyptian firms, similar in magnitude to the effects estimated for U.S. exports. This pattern suggests that firms did not simply reallocate exports from non-U.S. destinations to the U.S. after treatment, since exports to both destinations increase. By exporting under the QIZ agreement, firms expand their exports to both the U.S. and non-U.S. markets, pointing to some underlying productivity gain such as a decrease in fixed costs, a learning by doing effect in export activities, better matching between suppliers and buyers, and/or economies of scale. Figure A4 in Appendix A, estimates the PPML version of this regression, and shows very similar results.

Number of Destinations

I next examine whether the QIZ agreement encouraged firms to increase the number of countries to which they export. Expanding the number of export destinations is often interpreted in the trade literature as a form of diversification and market upgrading. According to models of multi-destination exporting (Eaton, Kortum,

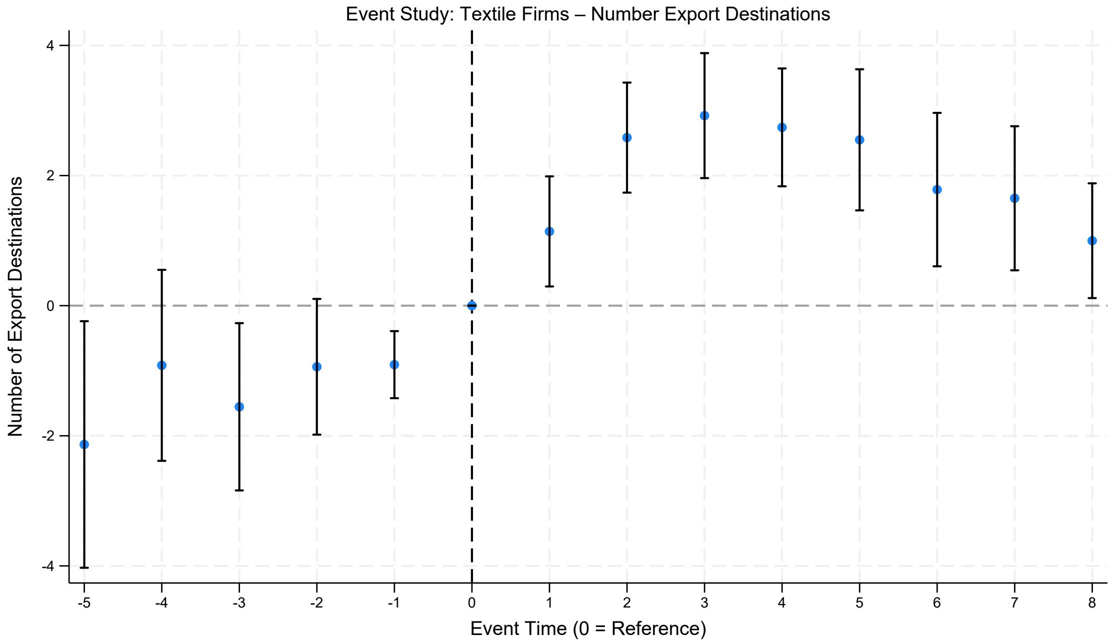


Figure 13: Event Study showing the dynamic effects of QIZs on the number of export destinations for Textile and Apparel Firms. Coefficients report OLS estimates using the number of destination countries as the dependent variable.

and Kramarz, 2011; Bernard et al., 2011), firms that face lower trade costs or improved access to buyer networks tend to enter new markets sequentially once initial fixed export costs are covered. Therefore, if the QIZ agreement reduced the costs of exporting or improved firms' credibility and relationships with foreign buyers, I would expect treated firms to expand not only their sales to the U.S. but also the set of other countries they serve. Figure 13 shows that the QIZ agreement drives treated firms to increase their number of export destinations by 3 and this persists 5 years after the treatment. Panels (a) in Figure A5 of Appendix A reports the results taking the log(x+1) of the number of destinations as the dependent variable while panel b reports the coefficients from a ppml regression. Both panels show similar results.

Number of Products

I next examine whether the QIZ agreement affected the range of products exported by treated firms. An increase in the number of exported products reflects product diversification and expansion along the extensive margin. The trade literature links such diversification to reductions in both variable and fixed export costs (Bernard, Redding, and Schott, 2010; Mayer, Melitz, and Ottaviano, 2014). By eliminating U.S. tariffs on qualifying exports, the QIZ agreement directly lowered variable trade

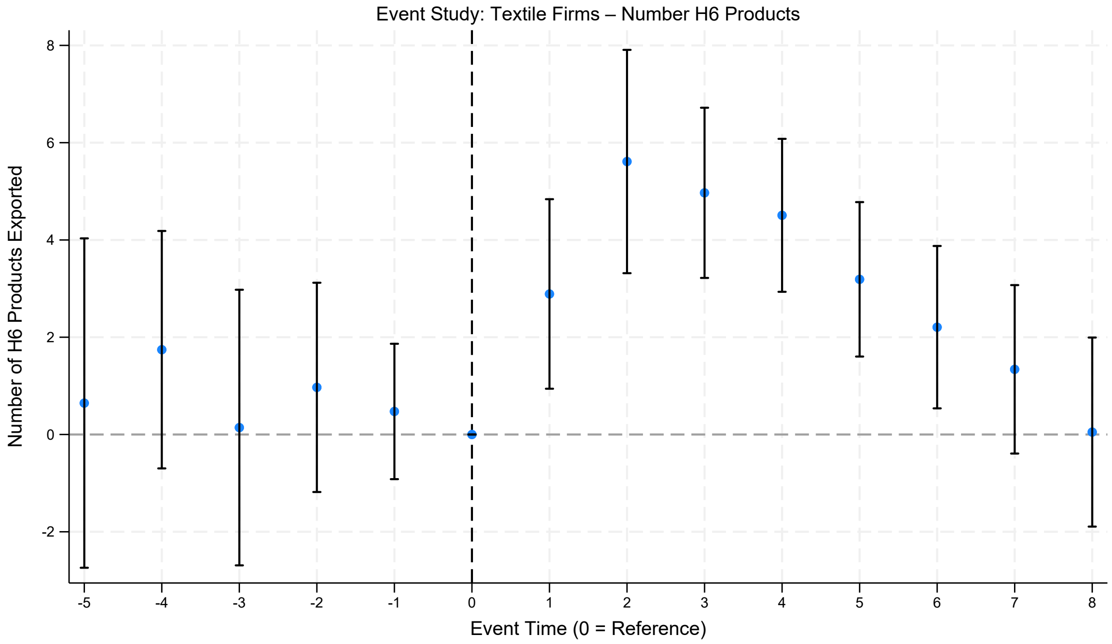


Figure 14: Event-study estimates of the dynamic effects of QIZs on the number of exported products for textile and apparel firms. Coefficients are OLS estimates using the number of exported HS product codes as the dependent variable.

costs, making it more profitable for firms to export additional products. At the same time, participation in the QIZ network may have reduced fixed export costs by improving access to imported inputs, easing compliance with export standards, and facilitating relationships with international buyers. Together, these effects could have encouraged treated firms to broaden their product range following treatment. Figure 14 shows that treated firms increased the number of products they exported by roughly six products relative to the control group, but this effect gradually declined and disappeared eight years after treatment. This pattern implies that the QIZ agreement led to a temporary diversification in firms' export portfolios, likely as firms took advantage of lower trade costs and new opportunities to experiment with additional product lines. Over time, however, firms appear to have refocused on their most profitable products, suggesting a process of rationalization consistent with models in which initial reductions in trade costs encourage product expansion, followed by specialization as competition intensifies and firms concentrate on their most profitable varieties.

Panels (a) in Figure A6 of Appendix A reports the results taking the log(x+1) of the number of destinations as the dependent variable while panel b reports the coefficients from a ppml regression. Both panels exhibit similar results.

6.4 Heterogeneous Effects

In this section, I investigate the heterogeneous effects of QIZ treatment on firms. First, I examine whether the increase in trade is driven primarily by incumbent firms or new entrants. Second, I analyze how these effects differ across successive QIZ expansion waves. Finally, I investigate how these effects differ across firm size as measured by their exports prior to their treatment.

Incumbents or New Entrants?

Determining whether QIZs expand trade with the U.S. by boosting the exports of incumbent firms (intensive margin) or by inducing new firms to start exporting (extensive margin) is important because these channels have different implications. The first suggests that the observed increase in exports under the QIZ agreement results from a reduction in variable trade costs (Bernard et al., 2003; Melitz, 2003), whereas the second points to a reduction in fixed export costs, which the trade literature identifies as a key driver of export entry and productivity growth (Das et al., 2007; Eaton et al., 2008).

I define new entrants as firms that do not export anything in 2005, the first year observed in our data. Conversely, incumbent firms are those that are observed in the dataset in 2005.

Figure 15 reports the results comparing coefficients for the full sample of textile firms, incumbents only, and new entrants only. As shown, the positive effects of QIZs on textile firms are almost entirely explained by incumbent firms, whose exports increase sharply after treatment. When restricting the sample of firms to incumbents or new entrants, pre-trends are absent, indicating the our inference still holds. In general, these results suggest that the effects are primarily driven by incumbent firms. This indicates that QIZ treatment operates mainly through a reduction in variable trade costs, which enhances the performance of existing exporters but may have negative distributional consequences, rather than through a reduction in fixed exporting costs that would allow new firms to enter and compete, potentially driving less productive firms out and increasing aggregate productivity in spirit of Melitz (2003).

Effects Across Early vs Late QIZ Expansions

In this subsection, I examine whether the increase in exports by treated firms varies by the wave of expansion. As shown in Figure 3, QIZ expansions occurred along two

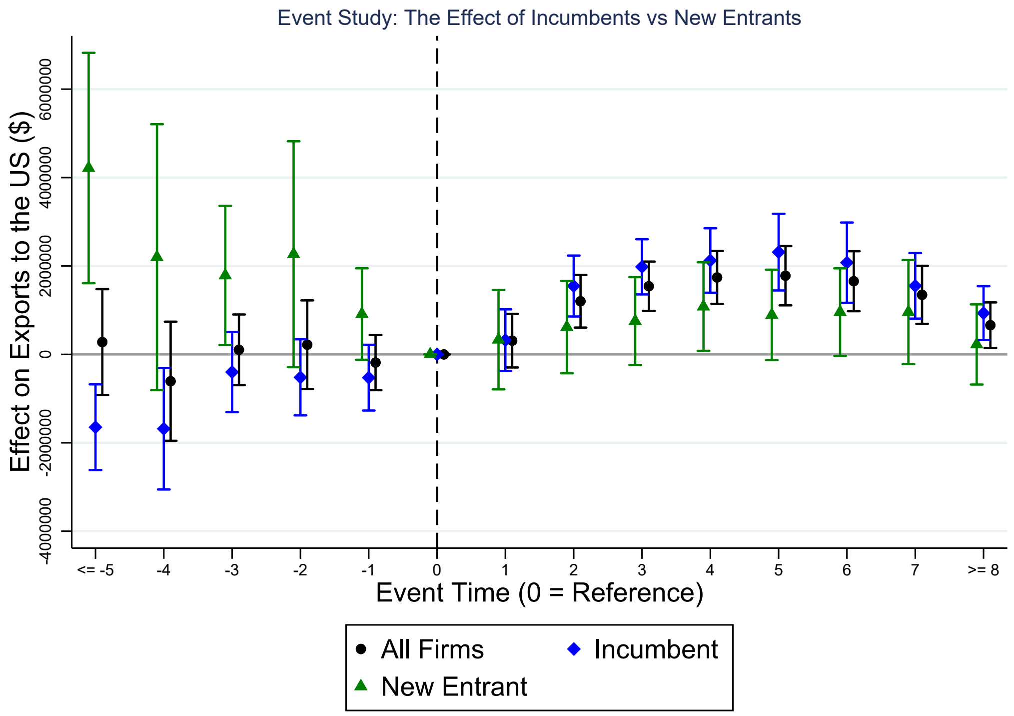


Figure 15: Event-study estimates of the dynamic effects of QIZs on exports to the United States for all, incumbent, and new-entrant textile and apparel firms. Coefficients are OLS estimates using export values as the dependent variable.

dimensions: geographically and over time. Since our customs data do not include firm location, I leverage the fact that expansions unfolded in a staggered manner across time. I classify firms in two groups: early-treated firms, which were treated in 2009 or earlier, and late-treated firms, which were treated after 2009. Firms treated earlier are likely to have been located in the regions designated as QIZs in 2004 and 2005, which were the areas where the textile and apparel industry was concentrated prior to the QIZ agreement. In contrast, later-treated firms could also be located in regions designated as QIZs in 2009 and 2013, which are generally less developed (CAPMAS, 2017) and have lower levels of manufacturing activity and a smaller textile industry.

Figure 16 reports the results comparing coefficients for three samples: the full set of textile firms, the sample excluding late-treated firms, and the sample excluding early-treated firms. As shown, the positive effects of QIZs on textile firms are driven entirely by early-treated firms, with little impact observed for later-treated firms. This pattern likely reflects the fact that early-treated firms were concentrated in regions with pre-existing textile and apparel clusters, better infrastructure, and established supply chains, allowing them to take advantage of the QIZ agreement more effectively. Combined with our earlier finding that incumbent firms drove the

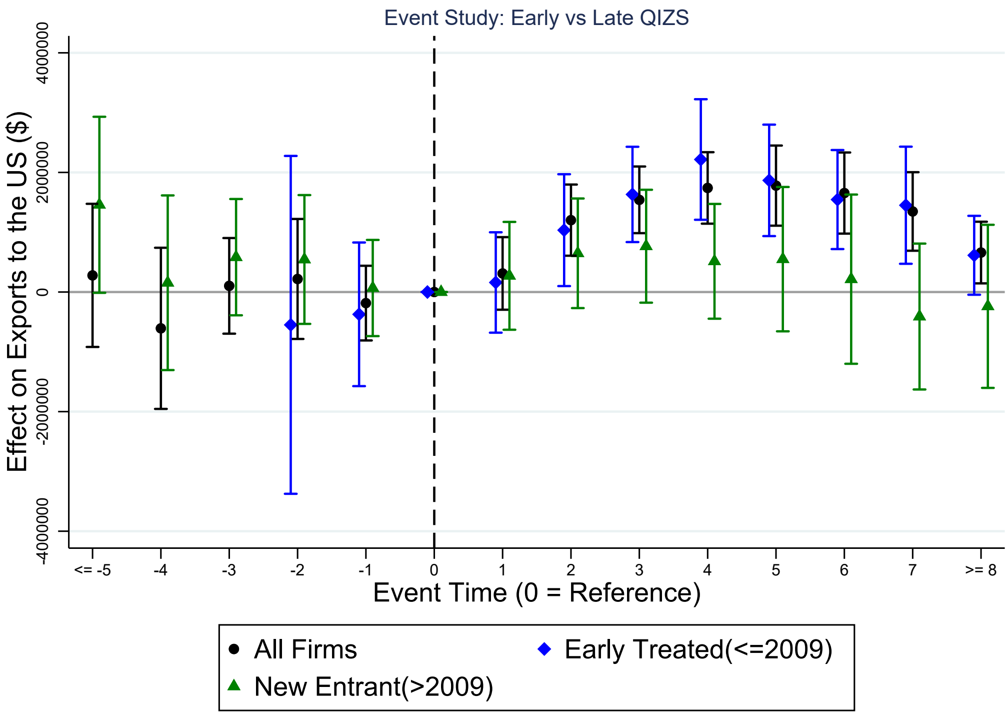


Figure 16: Event-study estimates of the dynamic effects of QIZs on exports to the United States for all, early-treated, and late-treated textile and apparel firms. Coefficients are OLS estimates using export values as the dependent variable.

overall results, this suggests that the benefits of QIZ treatment accrued primarily to firms that were already established and located in historically strong manufacturing regions, rather than to new entrants or firms in less developed areas.

Effects Across Firms by Size

Finally, in this suubsection, I examine whether the increase in exports by treated firms varies by firm size, as measured by their average value of exports before their treatment. I classify firms in two groups based on the average value of exports that treated firms had before their treatment and that untreated firms had over the entire sample. Hence big firms are defined as firms whose value of exports pre-treatment was greater than this average, while small firms are those whose average pre-treatment exports were less than this average.

Figure 17 reports the results comparing coefficients for three samples: the full set of textile firms, the sample excluding big-treated firms, and the sample excluding small-treated firms. As shown, the positive effects of QIZs on textile firms are driven by both big and smaller firms. However the relative increase in exports to the U.S after the event is much greater for smaller firms and persists for a longer time. This pattern likely reflects the fact that smaller treated firms exported very little prior to

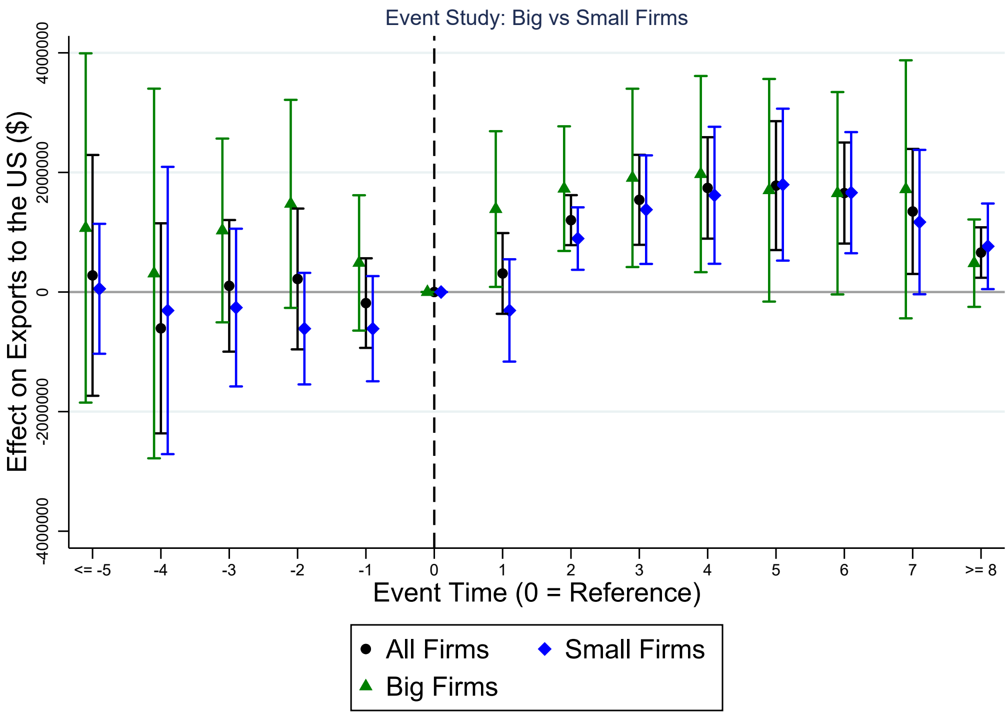


Figure 17: Event-study estimates of the dynamic effects of QIZs on exports to the United States for all, large, and small treated textile and apparel firms. Coefficients are OLS estimates using export values as the dependent variable.

the QIZ agreement, whereas bigger firms were already exporting to the world and possibly to the U.S as well through the Multi-fiber quota agreement.

Taken together, these results yield a coherent story. First, the QIZ effects are driven primarily by incumbent firms rather than by new entrants. Second, they are concentrated among early-treated firms—likely those embedded in pre-existing textile clusters with better infrastructure and buyer linkages. Third, when splitting by size, both small and large firms benefit: the treatment raises exports for each group. The magnitudes, however, differ in intuitive ways-relative gains are larger and more persistent for smaller firms which started from lower pre-QIZ export levels, while absolute gains are naturally larger for big firms given their greater scale.

Combined with our earlier event-study evidence, this pattern suggests that QIZs primarily operated through reductions in variable trade costs and coordination frictions for already-active exporters, with complementary advantages in regions that were designated early and already had textile capacity. Rather than inducing widespread de new entry, the agreement disproportionately amplified the performance of incumbent firms across the size distribution, while delivering especially large percentage increases for smaller incumbents.

7 Firm-Level Analysis from Censuses

In this section, I use two waves of the Egyptian Firm-Level Industrial Census (2012 and 2017) to estimate equation (3) and examine the impact of a region's designation as a QIZ on textile firm outcomes, including employment and wages. The identification strategy leverages the 2013 QIZ expansion as the treatment, excluding from the analysis all regions that had previously been designated as QIZs. This design allows me to compare textile firms located in newly designated QIZ regions with those in regions that were never designated. As discussed in the heterogeneity analysis, later-treated firms exhibited smaller export increases than earlier-treated firms; however, I still examine whether QIZ designation influenced textile firms at the regional level.

7.1 Pre-Treatment Balance Test

Since there is only one pre-treatment and one post-treatment observation, it is essential to verify that treatment and control observations were not statistically different in the pre-period. Otherwise, the DDD estimates could be biased and fail to capture the true effect of the treatment (Ortiz-Villavicencio and Sant'Anna, 2025). To assess this, I compare average outcomes in the pre-treatment year (2012) for textile firms in treated regions with two groups: textile firms in non-treated regions and all other firms in both treated and non-treated regions.

Figure 18 reports balance checks for differences in the following dimensions: firm characteristics (age, size, ownership), inputs (labor, materials, capital), finance (revenue, costs, assets), and tax and subsidy exposure. I find no statistically significant differences between treated and control firms in any of these dimensions, except for firm size, which is controlled for in the regressions. This supports the validity of the identification strategy by reducing concerns about systematic pre-treatment differences that could bias the estimated effects.

7.2 Results

I first examine the effect of QIZs on firm-level production outcomes. I find evidence that QIZ designation improved the performance of treated firms in regions added to the program in 2013. Specifically, the total revenue, production, and gross value added of these firms increased relative to controls. Figure 19 shows these results, with estimated log coefficients close to 1, implying an increase of more than 150 % in

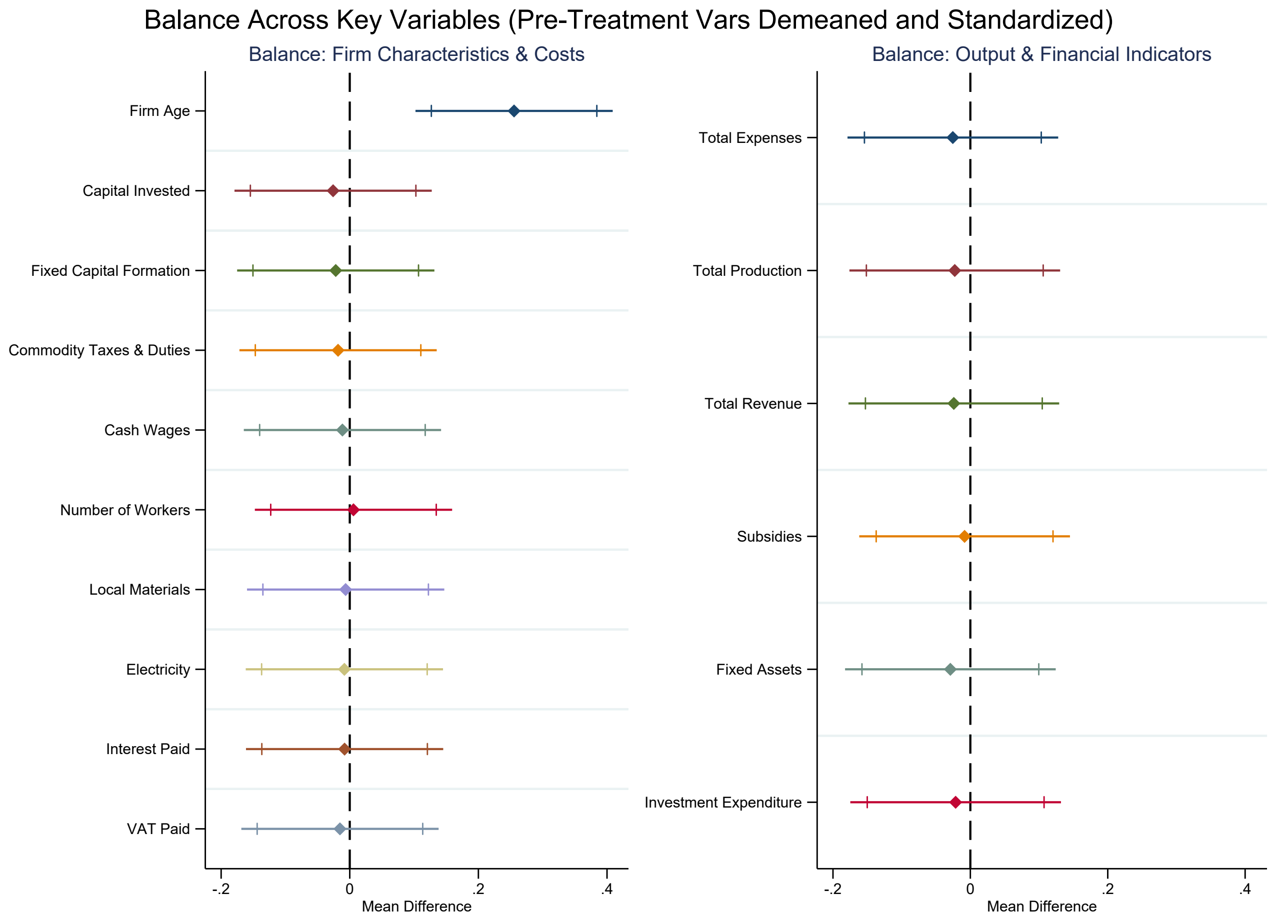


Figure 18: Pre-Treatment Period (2012) Balance Test Results

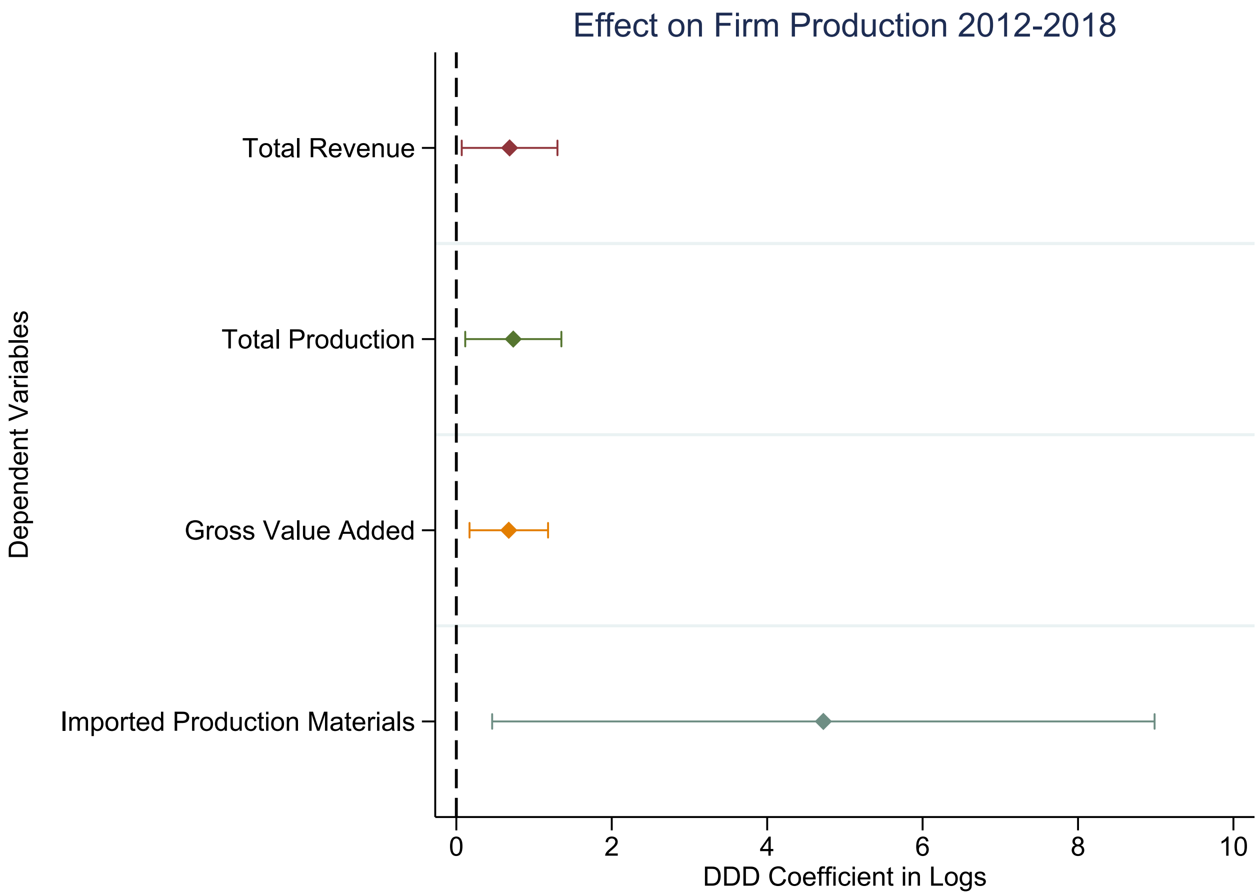


Figure 19: Firm Production Outcomes: DDD Estimates. Confidence intervals are shown at the 90% level.

these outcomes. Importantly, I also observe a substantial increase in the coefficient on imported production materials, which reinforces the validity of the empirical strategy: to qualify for QIZ benefits, firms must source a minimum share of inputs from Israel, making this an expected first stage channel of adjustment.

I then examine the effect of QIZs on firm-level employment. I find that QIZ designation increased the number of textile workers by roughly 100%, with most of this growth driven by male rather than female workers. When distinguishing between permanent and temporary employment, both categories show substantial increases. The gendered pattern of this effect is surprising, as globally the textile industry often absorbs female workers and serves as a key driver of structural change for women (Heath and Mobarak, 2015). However, this does not appear to be the case in Egypt, where male workers continue to dominate the textile industry, largely due to cultural and social norms (Bursztyn, González, and Yanagizawa-Drott, 2020; World Bank, 2023). Interestingly, both male and female unpaid workers decline following QIZ designation. This may reflect compliance with labor requirements tied

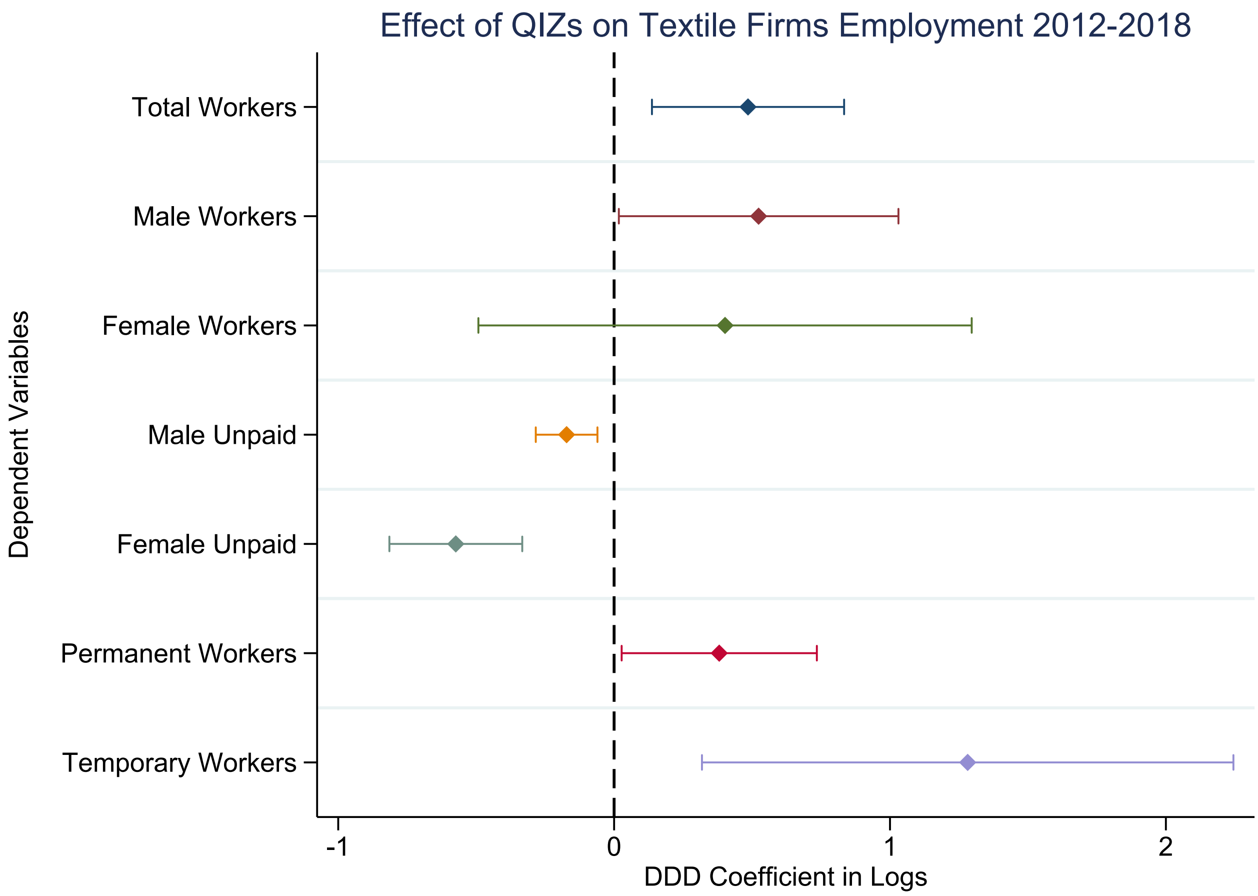


Figure 20: Firm Employment Outcomes: DDD Estimates. Confidence intervals are shown at the 90% level.

to QIZ participation, which likely incentivized firms to convert unpaid positions into temporary contracts. Overall, these findings suggest that while trade agreements such as the QIZ can boost employment, they may not produce the same gender-equalizing effects observed in other contexts.

Finally, we examine the effect of QIZ treatment on wages. As shown in Figure 21, we find no statistically significant impact on wages—whether measured in cash or in-kind compensation. This absence of wage growth, despite substantial increases in employment, may reflect labor market conditions in which an abundant supply of low-skilled workers prevents upward wage pressure, or the nature of job creation under QIZs, which often relies on short-term or lower-quality contracts rather than higher-paid employment. It may also reflect the fact that this analysis uses only the most recent QIZ expansion, which earlier results suggest had relatively limited effects on exports to the United States. In the next section, when examining both the 2009 and 2013 expansions jointly, we observe stronger wage responses.

Furthermore, these aggregate firm-level results may mask important gender-

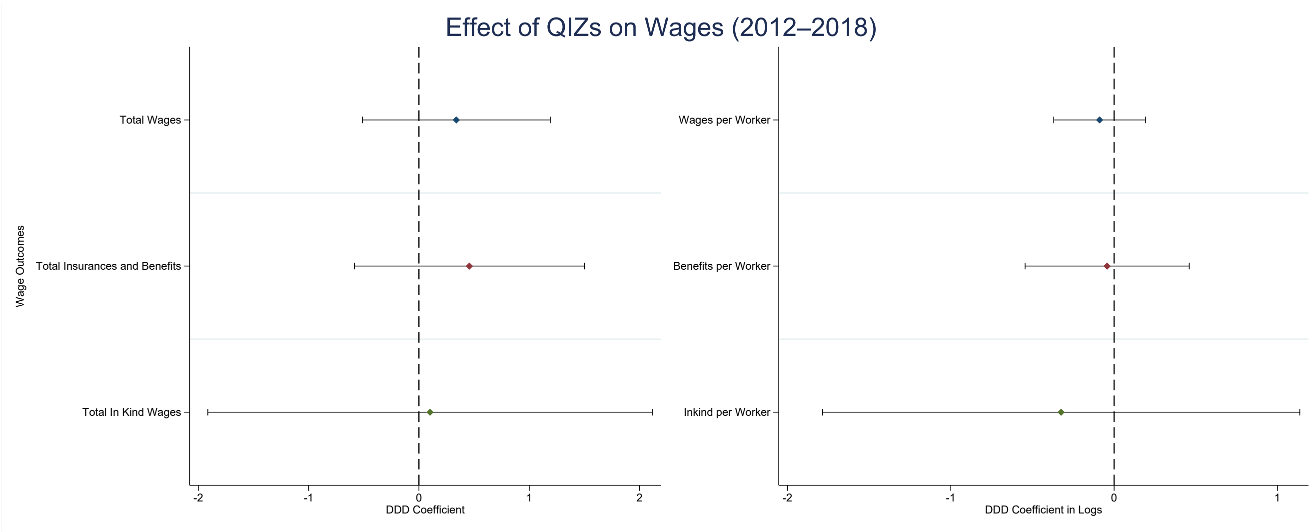


Figure 21: Firm Wages Outcomes: DDD Estimates. Confidence Intervals are at the 90% level

specific patterns that this dataset cannot capture, motivating the use of labor force survey data in the following section.

In Appendix A, we restrict the sample to manufacturing firms and repeat the analysis. The results are qualitatively similar, though estimated with less precision.

8 Labor Force Surveys

In this section, I use the Egyptian Labor Force Surveys from 2006 to 2017 to examine the effect of regions designated as QIZs on wages and social insurance coverage for workers in textile firms. I estimate equation (4) for this analysis. To validate the parallel trends assumption, I first present event study plots, and then I explore the aggregate results separately by gender.

Results

I begin by examining the effect of QIZ designation on the log wages of textile workers. The analysis exploits the 2009 and 2013 QIZ expansions and excludes workers in previously treated regions, for which pre-treatment data are unavailable. Figure 22 presents the corresponding event-study estimates. The pre-treatment coefficients are flat, supporting the parallel trends assumption. Although the post-treatment estimates are somewhat noisy, they indicate a positive and statistically significant increase in log wages following QIZ designation.

Next, I examine heterogeneity by gender. As shown in Table 1, QIZ designation leads to a positive and statistically significant increase in male wages. The esti-

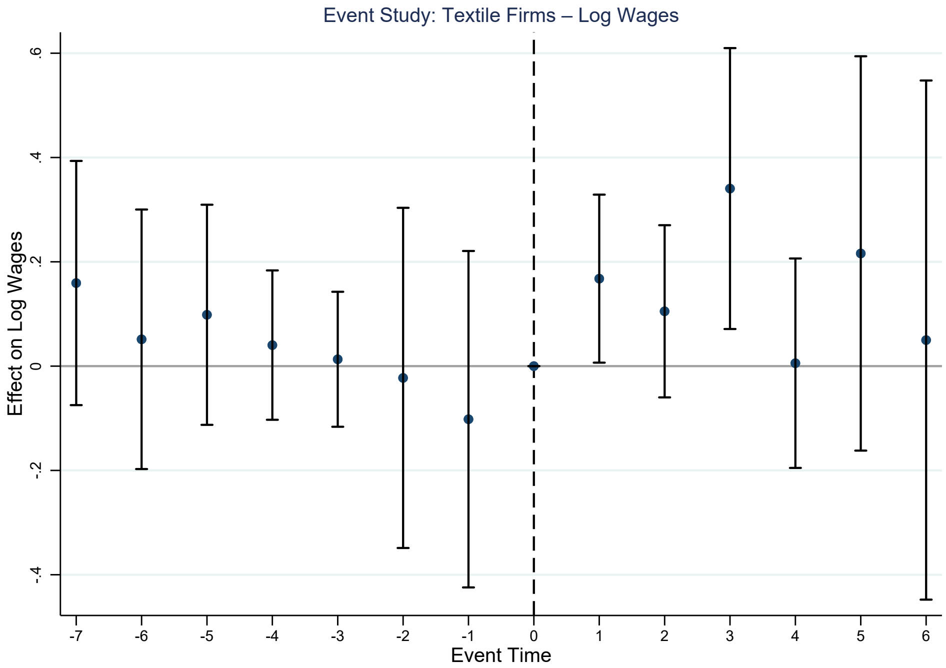


Figure 22: Worker Log Labor Wages: Dynamic Effects from DDD Estimates.Confidence Intervals are at the 95% level

Table 1: Impact of QIZ Designation on Log Wages by Gender

	(1) Log Wages	(2) Log Wages: Males	(3) Log Wages: Females
Textile×Gov×Post	0.134**	0.120**	0.491
	(0.050)	(0.057)	(0.322)
Observations	155,689	127,335	28,071
Adjusted R^2	0.528	0.532	0.564

Standard errors in parentheses. Sample excludes food and chemical activities

Weighted by pweight. Clustered standard errors at governorate level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

mated effect for female workers is large but imprecisely estimated and statistically insignificant, likely due to the small number of female textile workers in the data. This pattern is consistent with the firm-level evidence presented earlier, suggesting that male workers benefited more from QIZ participation than female workers.

I then examine the effect of QIZ designation on workers' social insurance status. Following the same approach as for wages, I first assess pre-trends and then estimate gender-specific effects. Figure 2 presents the event-study results. The pre-treatment coefficients show no detectable deviations from parallel trends, supporting the validity of the DDD design. The dynamic estimates reveal little evidence of a statistically significant or persistent increase in overall social insurance coverage following QIZ

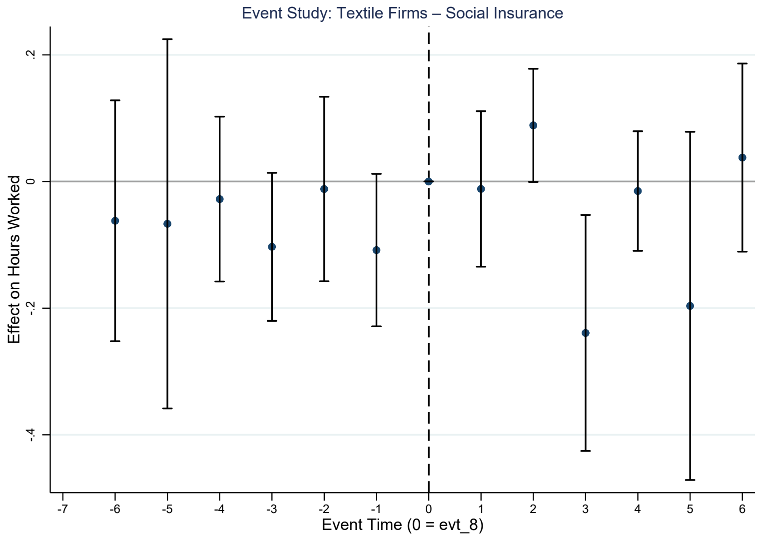


Figure 23: Worker Social Insurance Status: Dynamic Effects from DDD Estimates

designation. However, the gender-disaggregated estimates reveal a more nuanced pattern. As shown in Table 2, QIZ designation increases social insurance take-up among male workers in a statistically significant manner, whereas the corresponding effect for female workers is statistically insignificant. This result is consistent with earlier findings from both firm-level and worker-level analyses, and reinforces the interpretation that male workers benefited more from QIZ participation in the Egyptian context—particularly with respect to the formality of their employment.

Table 2: Impact of QIZ Designation on Social Insurance Take-Up by Gender

	(1) Social Insurance	(2) SI: Males	(3) SI: Females
Textile×Gov×Post	0.069*	0.097**	0.131
	(0.039)	(0.039)	(0.078)
Observations	394204	307710	86222
Adjusted R^2	0.997	0.995	0.999

Standard errors in parentheses. Sample excludes food and chemical activities.

Weighted by pweight. Standard errors clustered at governorate level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

9 Quantitative Model

This section develops a quantitative heterogeneous-firm trade model tailored to the Qualifying Industrial Zones (QIZ) setting. The model serves two purposes. First, it provides a disciplined mapping from the estimated firm-level responses—

export expansion, destination and product diversification, and evidence consistent with productivity gains—to structural mechanisms: endogenous export participation, endogenous compliance with rules-of-origin (ROO) through costly Israeli input sourcing, and endogenous productivity upgrading induced by U.S. market access. Second, it allows aggregation of these micro responses into economy-wide outcomes, including equilibrium wages and employment reallocation across locations, domestic price indices, and welfare. The model is calibrated to match moments from the customs microdata and is used to quantify the welfare effects of QIZ and to evaluate counterfactual ROO stringency by varying the Israeli content requirement.

9.1 Environment

Egypt consists of two regions $r \in \{Q, N\}$, where Q denotes QIZ-eligible locations and N denotes non-QIZ locations. Firms are region-fixed, while workers can move across regions. The economy produces differentiated varieties in multiple sectors indexed by $s \in S$ (e.g., textiles/apparel, food). Firms sell domestically ($j = \text{EG}$) and export to foreign destinations $j \in \{\text{US}, \text{RW}\}$ (rest of world).

Egypt is modeled as a small open economy relative to the U.S. and the rest of the world. In foreign destinations, sectoral expenditures E_{js} and sectoral price indices P_{js} are taken as exogenous. Israel enters through the ROO constraint: compliant QIZ exporters must source a minimum share of intermediates from Israel, which raises their variable input costs. Israeli intermediate input prices are treated as exogenous, consistent with the assumption that Egypt is small relative to Israeli intermediate markets at the relevant scale.

9.2 Preferences and demand

Consumers in each destination j have nested CES preferences over sectors and varieties within sectors. Upper-tier utility in j is

$$U_j = \left(\sum_{s \in S} \beta_s^{1/\eta} \, U_{js}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{4}$$

where $\beta_s > 0$ are sector weights and $\eta > 0$ is the elasticity of substitution across sectors. Within each sector s,

$$U_{js} = \left(\int_{\omega \in \Omega_{js}} q_{js}(\omega)^{\frac{\sigma_s - 1}{\sigma_s}} \, d\omega \right)^{\frac{\sigma_s}{\sigma_s - 1}}, \tag{5}$$

where $\sigma_s > 1$ is the within-sector elasticity of substitution and Ω_{js} is the set of varieties available in destination j.

Standard CES demand implies that, in destination j and sector s, demand for a variety priced at $p_{js}(\omega)$ is

$$q_{js}(\omega) = \left(\frac{p_{js}(\omega)}{P_{js}} \right)^{-\sigma_s} \frac{E_{js}}{P_{js}}, \tag{6}$$

where E_{js} is total expenditure on sector s in destination j, and P_{js} is the sectoral CES price index:

$$P_{js} = \left(\int_{\omega \in \Omega_{js}} p_{js}(\omega)^{1-\sigma_s} \, d\omega \right)^{\frac{1}{1-\sigma_s}}. \tag{7}$$

Revenue from sales to j can be written as

$$R_{js}(\omega) = p_{js}(\omega) q_{js}(\omega) = E_{js} \left(\frac{p_{js}(\omega)}{P_{js}} \right)^{1-\sigma_s}. \tag{8}$$

For Egypt ($j = \text{EG}$), expenditures and price indices are endogenous. Aggregate income in Egypt is

$$Y_{\text{EG}} = w_Q L_Q + w_N L_N + T, \tag{9}$$

where w_r and L_r are the wage and employment in region r and T is a net transfer that closes the small-open economy (calibrated to match baseline external imbalance). Sector expenditures satisfy the standard CES allocation

$$E_{\text{EG},s} = \beta_s \left(\frac{P_{\text{EG},s}}{P_{\text{EG}}} \right)^{1-\eta} Y_{\text{EG}}, \tag{10}$$

with aggregate price index

$$P_{\text{EG}} = \left(\sum_{s \in S} \beta_s P_{\text{EG},s}^{1-\eta} \right)^{\frac{1}{1-\eta}}. \tag{11}$$

9.3 Firms: entry, technology, and marginal costs

In each region r and sector s, a mass of potential entrants pays a sunk entry cost $w_r f_{rs}^E$ to draw productivity ϕ from a Pareto distribution:

$$G_s(\phi) = 1 - \left(\frac{\phi_{\min,s}}{\phi}\right)^{\theta_s}, \qquad \phi \geq \phi_{\min,s}, \tag{12}$$

with shape parameter $\theta_s > \sigma_s - 1$.

A firm with productivity ϕ produces output using labor l and a composite intermediate input m:

$$y = \phi\, l^{\alpha_s} m^{1-\alpha_s}, \tag{13}$$

where $\alpha_s \in (0,1)$ is the labor share. Cost minimization yields the marginal cost of production:

$$mc_{rs}(\phi; c_{m,s}) = \frac{1}{\phi} \cdot \frac{w_r^{\alpha_s} c_{m,s}^{1-\alpha_s}}{\alpha_s^{\alpha_s}(1-\alpha_s)^{1-\alpha_s}}, \tag{14}$$

where $c_{m,s}$ is the unit cost of the intermediate composite faced by the firm. The model distinguishes two sourcing regimes:

Noncompliance (free sourcing). Noncompliant firms source intermediates from the rest of the world at unit cost

$$c_{m,s}^N = p_{\mathrm{RW},s}. \tag{15}$$

Compliance (ROO sourcing). Compliant firms must satisfy a minimum Israeli-content requirement $\gamma_s \in (0,1)$, implemented as a Cobb–Douglas input bundle:

$$m = (m_{\mathrm{IL}})^{\gamma_s} (m_{\mathrm{RW}})^{1-\gamma_s}. \tag{16}$$

The corresponding unit cost index is

$$c_{m,s}^C(\gamma_s) = \frac{p_{\mathrm{IL},s}^{\gamma_s} p_{\mathrm{RW},s}^{1-\gamma_s}}{\gamma_s^{\gamma_s}(1-\gamma_s)^{1-\gamma_s}}, \tag{17}$$

and complying also requires paying a fixed compliance cost $w_Q f_s^C$.

9.4 Trade costs, tariffs, pricing, and export revenues

Shipping from region r to destination j in sector s incurs iceberg trade costs $d_{rjs} \geq 1$. In addition, exporting to the U.S. is subject to an MFN tariff t_s^{MFN} unless the firm complies with QIZ. Let the total delivered wedge be

$$\tau_{rjs} = d_{rjs} \times \begin{cases} 1 + t_s^{\text{MFN}}, & j = \text{US and noncomplier}, \\ 1, & j = \text{US and complier}, \\ 1, & j \in \{\text{EG}, \text{RW}\}. \end{cases} \tag{18}$$

Delivered marginal cost is $\widetilde{mc}_{rjs}(\phi) = \tau_{rjs}\, mc_{rs}(\phi; c_{m,s})$.

Firms compete under monopolistic competition and take E_{js} and P_{js} as given. Profit maximization under CES demand implies constant markup pricing:

$$p_{rjs}(\phi) = \mu_s\, \widetilde{mc}_{rjs}(\phi), \qquad \mu_s = \frac{\sigma_s}{\sigma_s - 1}. \tag{19}$$

Substituting (19) into (8), export revenue to destination j is

$$R_{rjs}(\phi) = E_{js} \left(\frac{\mu_s\, \widetilde{mc}_{rjs}(\phi)}{P_{js}} \right)^{1-\sigma_s}, \tag{20}$$

which is strictly increasing in productivity ϕ.

9.5 Export participation, selection, and cutoffs

Serving destination j requires paying a fixed cost $w_r f_{rjs}$. Given pricing above, variable profits from selling to j satisfy $\pi_{rjs}^{var}(\phi) = R_{rjs}(\phi)/\sigma_s$. Exporting is profitable iff

$$\frac{1}{\sigma_s} R_{rjs}(\phi) \geq w_r f_{rjs}. \tag{21}$$

This defines a unique productivity cutoff ϕ_{rjs}^* such that firms export to j iff $\phi \geq \phi_{rjs}^*$. The model therefore generates intensive-margin responses (higher exports among incumbents) and extensive-margin responses (changes in the set of exporters) as trade costs and tariffs change.

9.6 Endogenous ROO compliance (QIZ eligibility)

Only firms in region Q are eligible to comply and receive tariff-free U.S. access. A QIZ firm chooses compliance if the gain in operating profits from avoiding the MFN tariff exceeds the increase in intermediate costs and the compliance fixed cost. Let $\Pi^N_{Qs}(\phi)$ denote maximized operating profits under noncompliance and $\Pi^C_{Qs}(\phi;\gamma_s)$ denote profits under compliance when the content requirement is γ_s. The firm complies iff

$$\Pi^C_{Qs}(\phi;\gamma_s) - w_Q f^C_s \geq \Pi^N_{Qs}(\phi). \tag{22}$$

Because compliance raises variable costs through $c^C_{m,s}(\gamma_s)$ but lowers the U.S. delivered wedge by removing t^{MFN}_s, compliance is more attractive when MFN tariffs are high and when the firm is sufficiently productive. This generates sector-specific and firm-specific take-up consistent with the empirical concentration of QIZ use in textiles/apparel.

9.7 Productivity upgrading and export spillovers

The model allows exporting (in particular, U.S. market access) to induce productivity upgrading. Upgrading scales productivity by $\delta_s > 1$ at fixed cost $w_r f^U_s$, so that $\phi' = \delta_s \phi$. Let $\Pi_{rs}(\phi)$ denote maximized operating profits (net of export fixed costs and, if applicable, compliance fixed costs) at productivity ϕ. The firm upgrades iff

$$\Pi_{rs}(\delta_s\phi) - \Pi_{rs}(\phi) \geq w_r f^U_s. \tag{23}$$

Upgrading lowers marginal costs for *all* destinations, and therefore increases exports not only to the U.S. but also to non-U.S. destinations. Under CES and constant markups, holding destination-level shifters fixed, revenues scale with productivity as $R \propto \phi^{\sigma_s - 1}$. Thus, for an incumbent exporter,

$$\frac{R'_{rjs}}{R_{rjs}} = \delta_s^{\sigma_s - 1}. \tag{24}$$

This mapping motivates the empirical use of export growth to non-U.S. destinations as a disciplined moment to identify upgrading, since it is not mechanically driven by the U.S. tariff preference.

9.8 Labor mobility and within-Egypt general equilibrium

Total labor supply is L. Workers choose region based on real wages. A parsimonious logit mobility structure implies regional labor shares

$$\lambda_r = \frac{(w_r/P_{\mathrm{EG}})^{\kappa}}{\sum_{r' \in \{Q,N\}} (w_{r'}/P_{\mathrm{EG}})^{\kappa}}, \qquad L_r = \lambda_r L, \tag{25}$$

where $\kappa > 0$ governs the responsiveness of migration to real wage differences. Regional labor-market clearing requires that labor supply equals labor demand from variable production and fixed costs (entry, exporting, compliance, upgrading), aggregated over firms:

$$L_r = \sum_{s \in S} M_{rs} \int l_{rs}(\phi)\, dG_s(\phi) + L_r^{fix}. \tag{26}$$

Domestic price indices $P_{\mathrm{EG},s}$ are determined by the set of varieties available in Egypt and their prices.

9.9 Equilibrium

An equilibrium is a set of wages $\{w_r\}$, labor allocations $\{L_r\}$, masses of entrants $\{M_{rs}\}$, export cutoffs $\{\phi^*_{rjs}\}$, compliance and upgrading decisions, and domestic price indices such that: (i) households optimize given prices and income, (ii) firms optimize input use, pricing, exporting, compliance, and upgrading, (iii) free entry holds in each (r, s), (iv) labor mobility (25) holds, (v) regional labor markets clear (26), and (vi) domestic price indices are consistent with CES aggregation. Foreign sectoral shifters (E_{js}, P_{js}) for $j \in \{\mathrm{US}, \mathrm{RW}\}$ and intermediate prices $(p_{\mathrm{IL},s}, p_{\mathrm{RW},s})$ are taken as given.

9.10 What the model will be used for: aggregate welfare and counterfactuals

The reduced-form evidence establishes large and persistent firm- and worker-level responses to QIZ exposure. The model is used to translate these micro responses into aggregate implications and to evaluate alternative policy designs that are not directly identified from the rollout.

(i) Welfare effects of QIZ. I compute the welfare effect of QIZ by comparing the equilibrium under the observed policy environment to a counterfactual without preferential access. In the baseline equilibrium, QIZ-eligible firms may comply with ROO and thereby obtain tariff-free U.S. access. In the counterfactual, QIZ preferences are removed (equivalently, all firms exporting to the U.S. face MFN tariffs), while all other parameters and foreign shifters are held fixed. Welfare in Egypt is measured as real income,

$$W \equiv \frac{Y_{\text{EG}}}{P_{\text{EG}}}, \tag{27}$$

and I also report distributional outcomes across regions, including real wages w_r/P_{EG} and employment shares L_r/L. This exercise quantifies how much of the policy's aggregate impact reflects (a) selection into exporting, (b) endogenous compliance costs through higher intermediate input prices, (c) induced upgrading, and (d) equilibrium wage and reallocation effects within Egypt.

(ii) Counterfactual ROO stringency: varying the Israeli content requirement. A central design parameter of QIZ is the Israeli content requirement γ_s. To evaluate the trade-off between conditionality and gains from market access, I solve the model repeatedly for a sequence of counterfactual requirements $\gamma_s \in [0, \bar{\gamma}_s]$, where $\bar{\gamma}_s$ is the baseline requirement in the agreement. Setting $\gamma_s = 0$ corresponds to removing the sourcing constraint while preserving preferential access, and increasing γ_s raises the variable cost of compliance through (17). For each value of γ_s, the model delivers predictions for compliance rates, the composition of exporters, exports to the U.S. and to non-U.S. destinations, Israeli intermediate imports, regional wages and employment, and welfare. Plotting $W(\gamma_s)$ traces a policy frontier: how the aggregate gains from preferential access depend on the degree of mandated integration with Israeli supply chains.

(iii) Additional counterfactuals and decomposition. The framework also supports decompositions that isolate mechanisms. For example, holding upgrading fixed (setting $\delta_s = 1$ or setting upgrading costs prohibitively high) isolates pure market-access and selection effects; holding compliance costs fixed isolates the role of ROO-induced variable costs; and turning off labor mobility isolates the importance of within-Egypt reallocation. These exercises clarify which margins are quantitatively most important for aggregate welfare and which margins drive the observed heterogeneity across firms and regions.

Calibration overview. The model parameters are calibrated to match moments from the data. Sector elasticities and tariffs are taken from external sources, while the Pareto shape, fixed costs, compliance costs, and upgrading parameters are disciplined by micro moments including exporter shares, export premia, changes in destination scope, changes in Israeli intermediate sourcing, and the magnitude of export spillovers to non-U.S. destinations.

10 Discussion

The experience of Egypt's QIZ program offers important lessons for designing trade agreements in geopolitically sensitive contexts. First, agreements that include politically motivated sourcing requirements, such as mandatory imports from a specific partner country, may help achieve short-term diplomatic goals but if not designed carefully, fail to generate lasting regional economic integration. The case of Jordan illustrates this. Once the broader United States–Jordan free trade agreement came into force, firms quickly stopped importing from Israel. This pattern highlights that such requirements do not shift underlying political or cultural attitudes but are directly tied to their economic benefits.

In the case of Egypt and Jordan, the failure to produce deeper regional cooperation cannot be separated from the unresolved nature of the Palestinian cause. Trade incentives alone cannot overcome this unresolved conflict and the deep cultural scars it leaves within the people of Jordan and Egypt. Policymakers in Europe considering similar approaches in regions with longstanding conflicts should recognize that integration cannot be achieved through trade provisions alone. Instead, success requires directly confronting structural political barriers.

Second, the sequencing and geographic targeting of trade programs influence how their benefits are distributed. My results show that the positive effects of QIZs on exports were concentrated among firms located in early-designated regions and among incumbent firms, while later expansions generated weaker impacts. Both large and small firms increased their exports, but smaller incumbents exhibited larger relative gains, suggesting that the program primarily benefited firms already integrated into global value chains. In addition to higher exports to the United States, treated firms expanded sales to non-U.S. destinations and showed short-term productivity improvements. Firms also temporarily increased the number of products and destinations they served before refocusing on their core lines, indicating an initial diversification followed by specialization as competition intensified.

At the firm level, QIZ designation increased revenues, production, and value added, but its effects on workers were uneven. Employment rose substantially, especially for male workers, and this expansion was accompanied by wage gains. By contrast, average firm-level wages showed no strong response, which reflects the concentration of new job creation in lower-paid or newly hired workers.

These patterns highlight an important trade-off in the design of geographically targeted trade policies. Prioritizing already strong industrial clusters can maximize initial export growth, but it may also deepen existing regional and social inequalities if benefits concentrate in more advanced areas and among specific worker groups. Incorporating less developed regions earlier in the process can encourage firms to invest in new locations, generating broader economic spillovers and promoting more inclusive development. Complementary place-based policies, such as infrastructure improvements, the creation of industrial zones, and targeted workforce training, are essential to ensure that trade liberalization supports balanced regional growth rather than reinforcing existing disparities (Kline and Moretti, 2014).

Finally, trade agreements must be designed with gender sensitivity in mind. The empirical analyses indicate that while QIZ participation increased male employment and social insurance coverage, female workers did not experience comparable gains. In contexts where social norms limit women's participation in the labor market, expanding trade alone might not lead to inclusive employment growth.

11 Conclusion

This paper examined the effects of Egypt's QIZ agreement on production, exports, and employment. The results demonstrate that the program generated substantial productivity-related gains, particularly among incumbent and early-treated textile and apparel firms located in established industrial clusters. These firms expanded their export volumes, diversified across products and destinations, and increased output, value added, and overall firm size. The pattern of effects is consistent with learning-by-exporting, incumbency advantages, and scale-related productivity improvements. Later-treated firms also benefited, although to a lesser degree. On the labor side, the program led to sizable employment growth, especially among men, reflecting pre-existing gender patterns in Egypt's manufacturing sector.

Despite these positive impacts, the broader distribution of gains was uneven across firms, regions, and workers. The evidence shows that QIZ-driven integration was concentrated within strong industrial clusters and did not generate deeper or more persistent forms of regional integration. This is consistent with international experience, where sourcing requirements tend to affect firms' short-run input choices without altering long-term economic linkages. Taken together, the findings suggest that trade agreements with conditional input-sourcing rules can successfully stimulate export growth and productivity improvements in targeted sectors, but their broader developmental impact depends critically on timing, geographic targeting, and complementary policies that support more inclusive and sustained industrial expansion.

An important next step is to translate these firm- and worker-level effects into aggregate and welfare implications. I therefore develop a quantitative heterogeneous-firm trade model with endogenous QIZ compliance, rules-of-origin costs, and productivity upgrading, disciplined by the micro evidence in this paper. The model will be used to quantify the economy-wide welfare effects of the program and to evaluate counterfactual designs that vary the Israeli content requirement, thereby characterizing the trade-off between conditionality and gains from market access.

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Appendix

A Figures

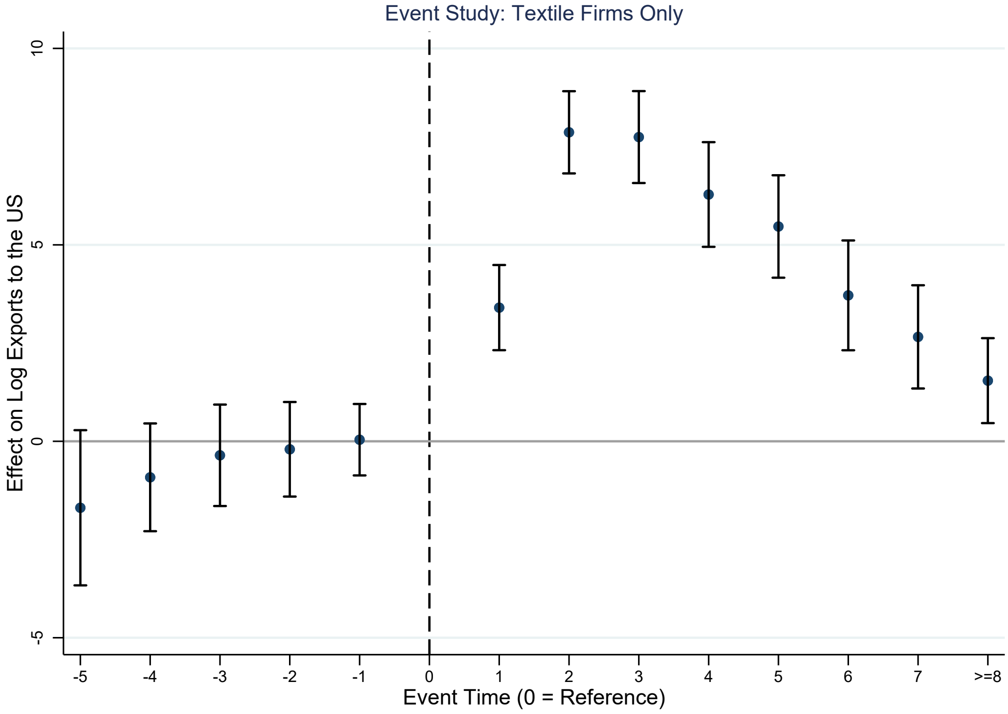
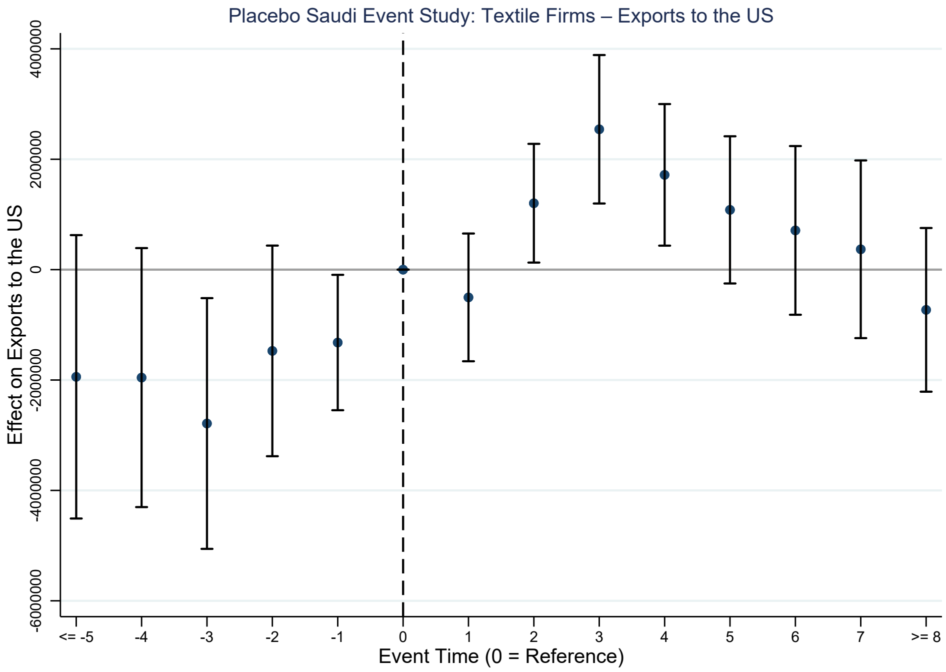
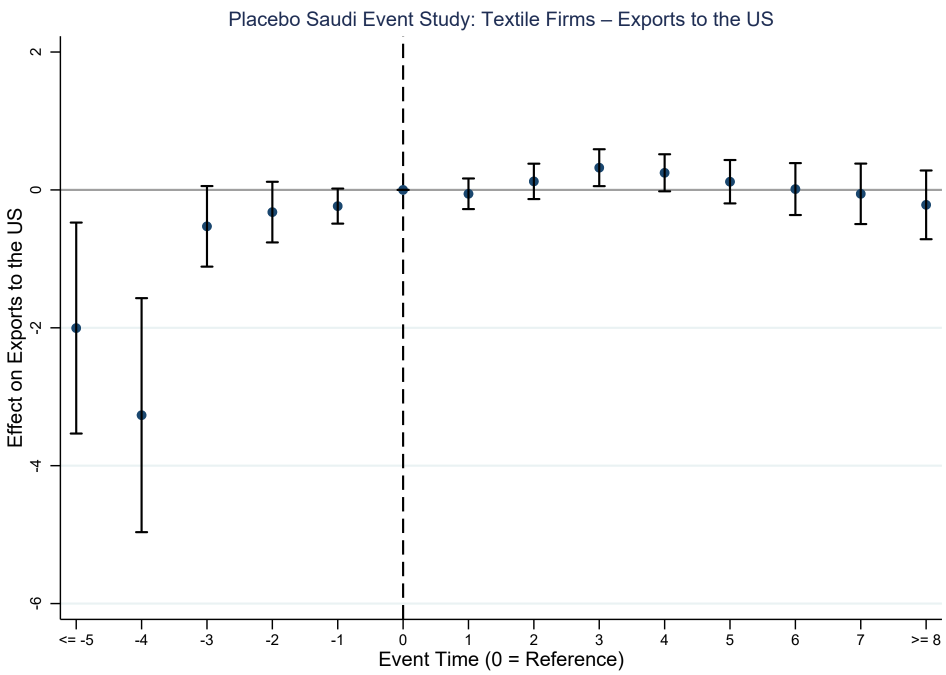


Figure A1: Event Study showing the dynamic effects of QIZs on the log(USExports +1) for Textile and Apparel Firms



(a) OLS



(b) PPML

Figure A2: Event study estimates using Saudi Arabia as a placebo country. Panel (a) reports OLS estimates using export values as the dependent variable, while Panel (b) reports PPML estimates of the same specification. The placebo test checks whether similar patterns would appear in a country not affected by QIZ treatment."

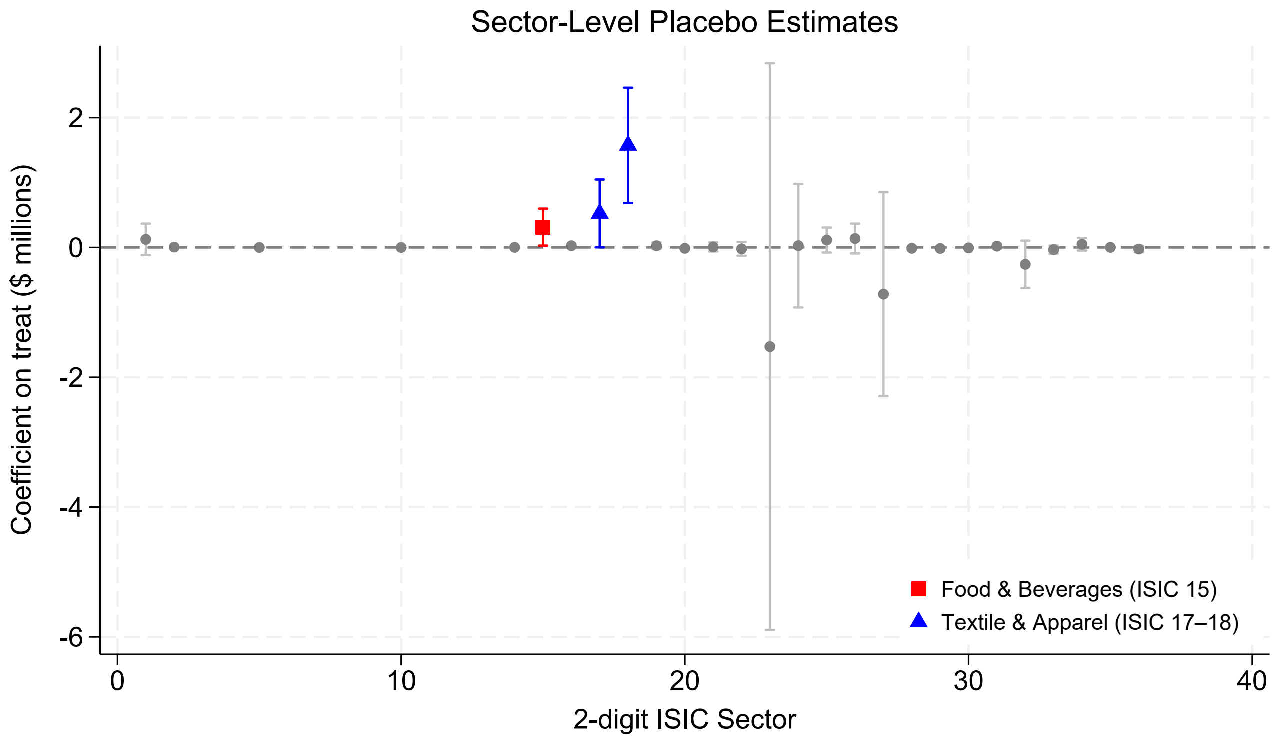


Figure A3: Figure showing the average treatment effects of QIZs on exports for each sectors. Coefficients reports OLS estimates using export values as the dependent variable.

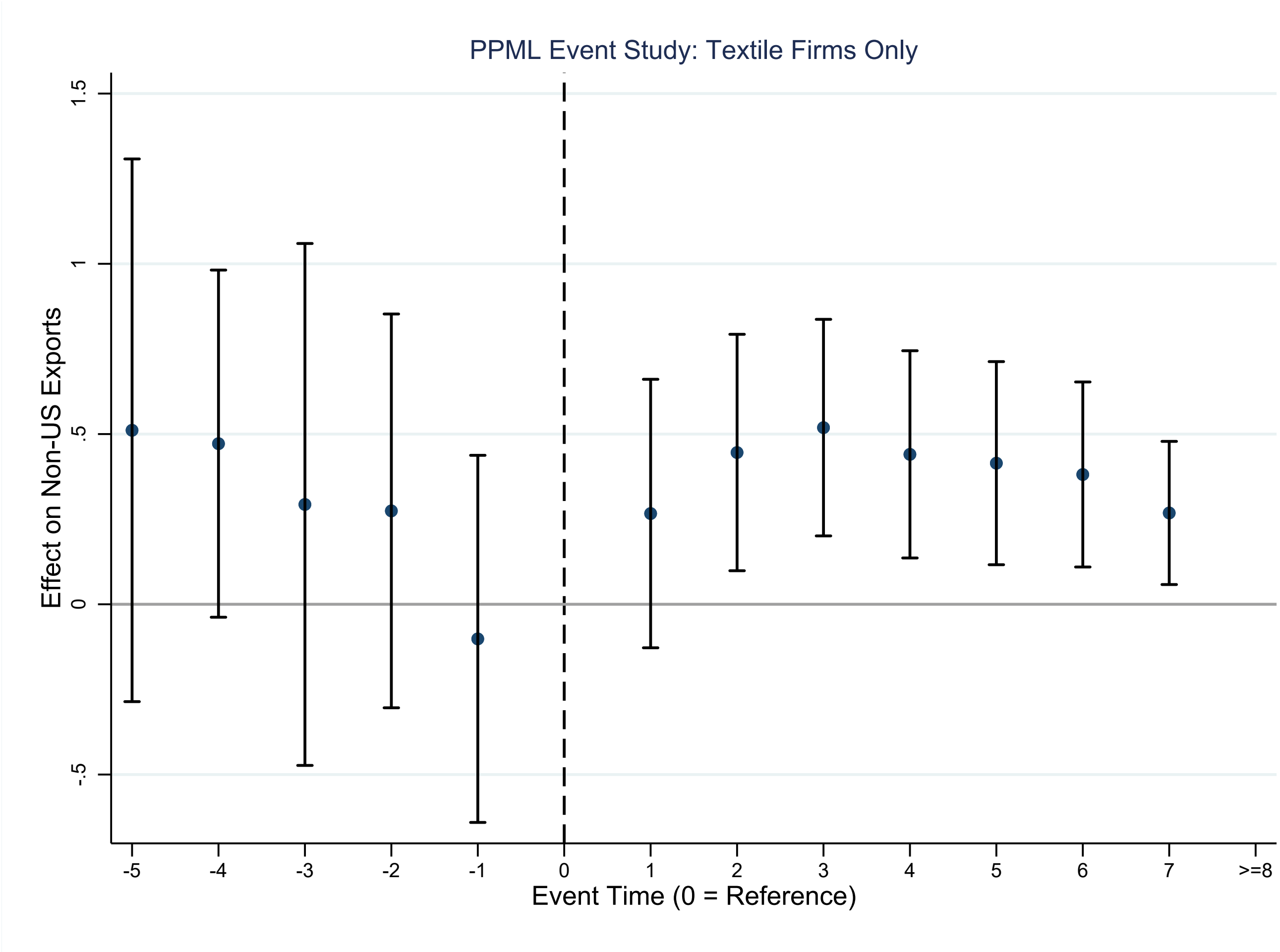
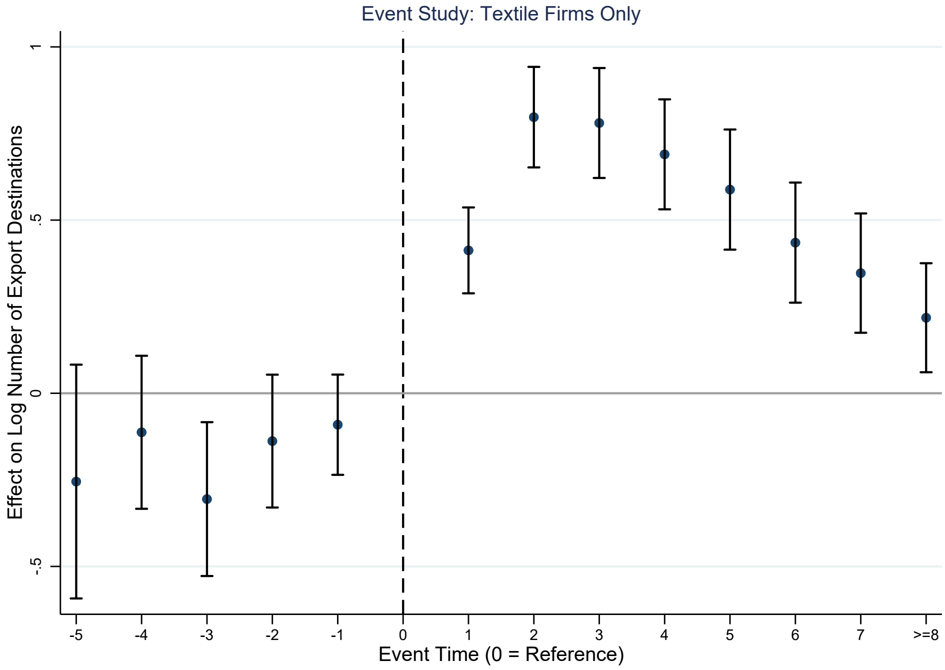
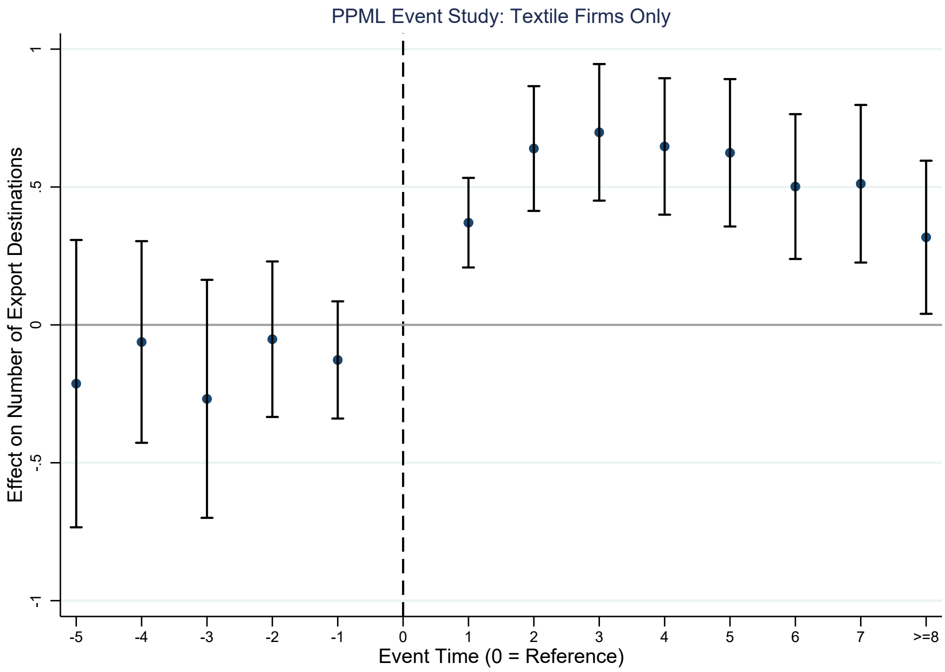


Figure A4: Event Study showing the dynamic effects of QIZs on exports to all Non-US destinations for Textile and Apparel Firms. Coefficients reports PPML estimates using export values as the dependent variable.

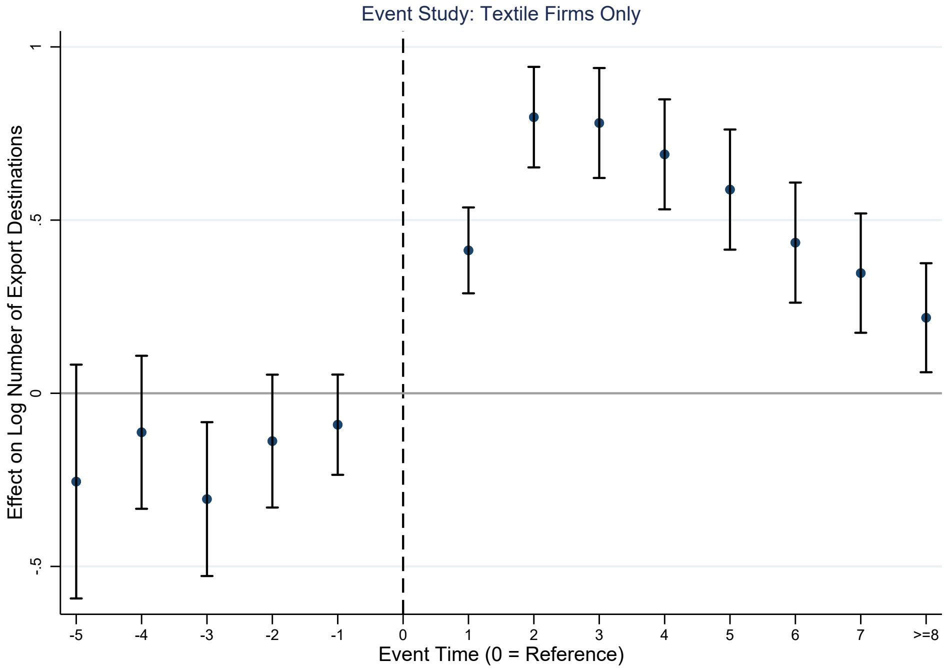


(a) OLS with log(numdestinations +1)

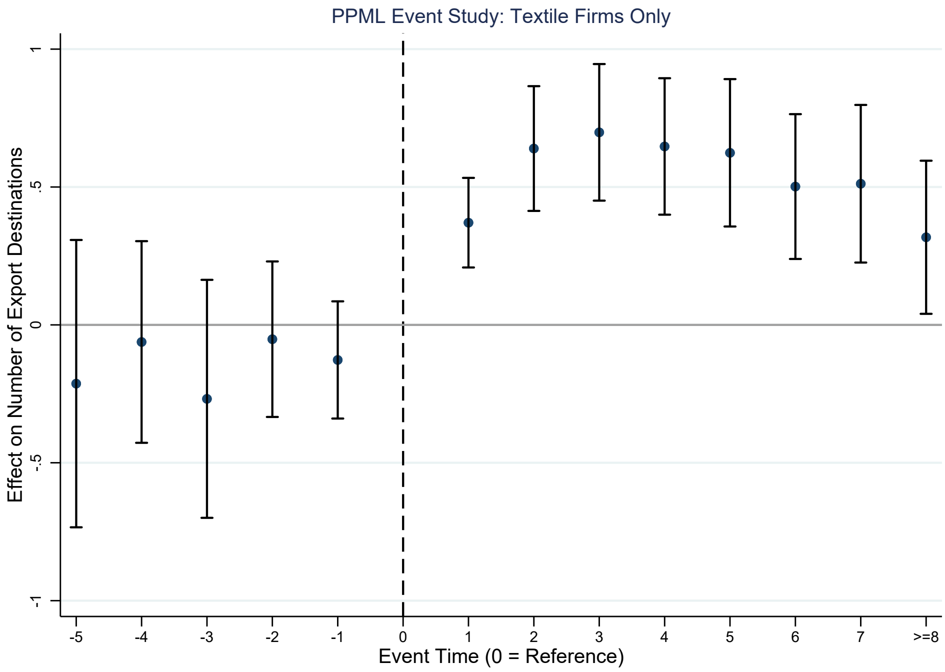


(b) PPML

Figure A5: Event study estimates with number of export destination countries as the dependent variable. Panel (a) reports OLS estimates using the log(x+1) of the number of export destinations as the dependent variable, while Panel (b) reports PPML estimates of the same specification.



(a) OLS with log(numproducts +1)



(b) PPML

Figure A6: Event study estimates with number of products exported as the dependent variable. Panel (a) reports OLS estimates using the log(x+1) of the number of products exported as the dependent variable, while Panel (b) reports PPML estimates of the same specification.

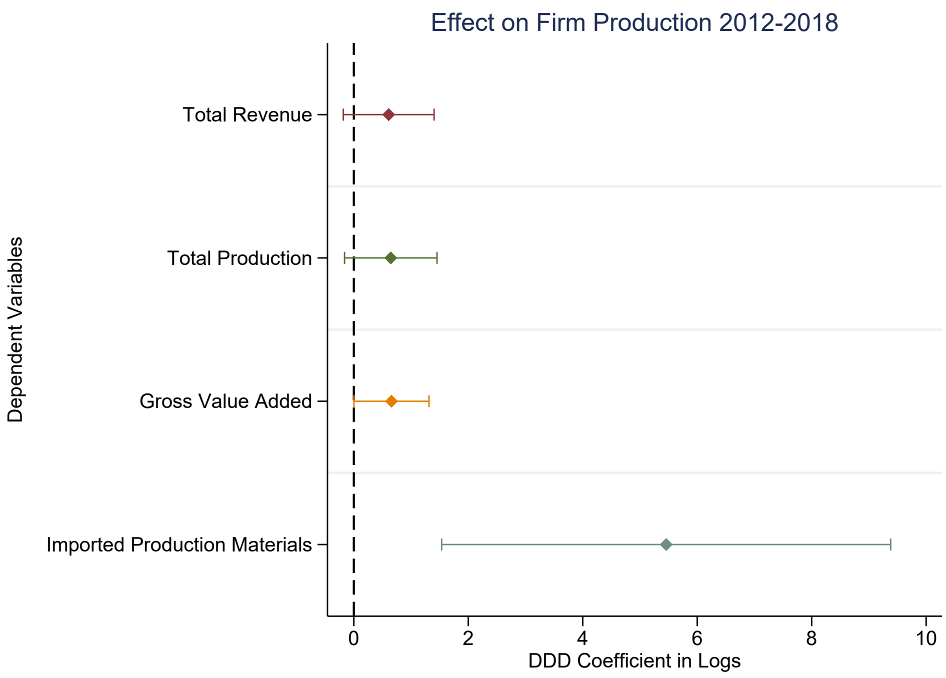


Figure A7: Firm Production Outcomes: DDD Estimates. Sample is restricted to manufacturing firms only. Confidence Intervals are at the 90% level

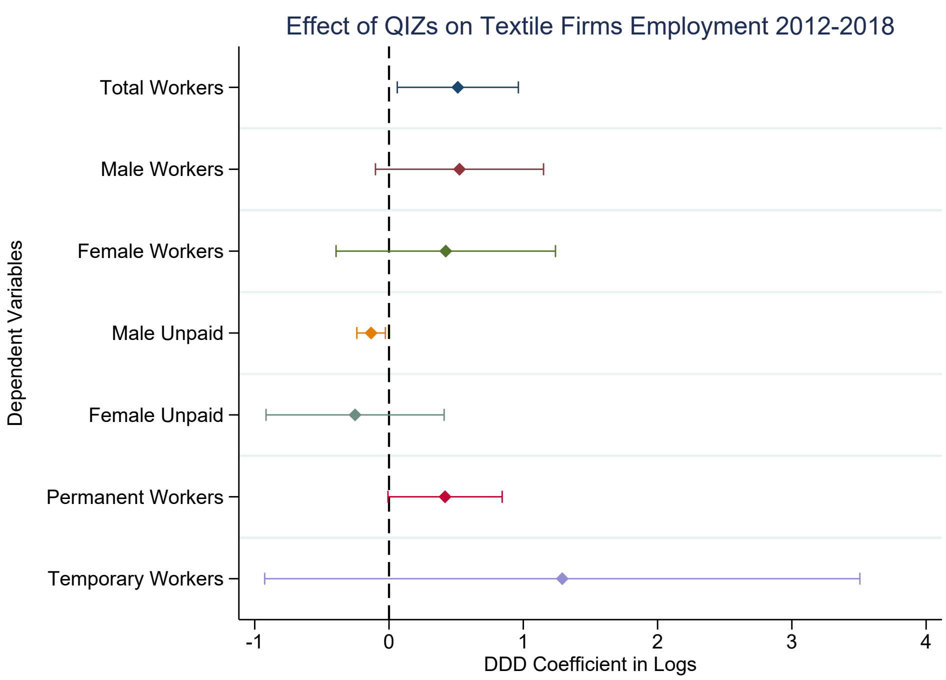


Figure A8: Firm Employment Outcomes: DDD Estimates. Sample is restricted to manufacturing firms only.Confidence Intervals are at the 90% level

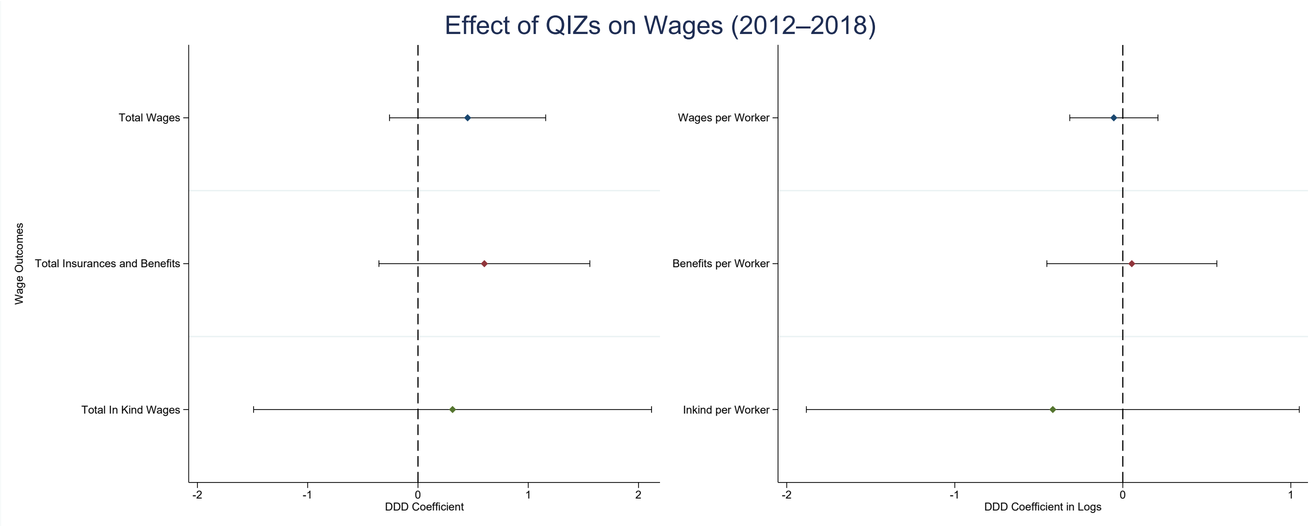


Figure A9: Firm Wages Outcomes: DDD Estimates. Sample is restricted to manufacturing firms only.Confidence Intervals are at the 90% level